

TOPOLOGICAL SEMANTICS FOR SCOPED COMPUTATIONAL PATHS

ARTHUR FREITAS RAMOS, RUY J. G. B. DE QUEIROZ,
ANJOLINA GRISI DE OLIVEIRA, AND TIAGO M. L. DE VERAS

ABSTRACT. Which computational distinctions survive a quotient topology? We study geometric computational paths modulo the rewrites declared by a presentation, comparing a trace-sensitive topology with one that observes endpoints, length, exact realization, and a coherent geometric representative. Duplicate labels separate these topologies. Conservative abbreviations can erase observable distinctions. A complete discrete-label presentation over the harmonic archipelago shows that geometric completeness alone does not make them agree. Equal-slot and binary timing can also give nonhomeomorphic observable based quotients. Continuous primitive word substitutions that are inverse up to scoped rewriting preserve the flat full-word global quotients and their final composable-pair domains.

The scoped quotient has continuous multiplication on the final quotient of composable representatives. We characterize agreement of this domain with the ordinary pullback of quotient arrows; the conditions imply ordinary multiplication is continuous. An open arrow projection is sufficient. For the universal presentation, a published open-map theorem supplies this condition under local path-connectedness and semilocal simple connectivity. Normal forms prove completeness for finite circle and torus presentations, and primitive interchange supports a product construction. An actual two-circle rose and its Cayley-edge covering extend represented-loop completeness to a nonabelian example.

Lean retains selected results in the binary model and separately checks the equal-slot model. Structural normalization and continuous reconstruction on composable word strata identify its literal-word and tree quotients, including global arrows and final composable-pair domains. A verification map records the models, assumptions, and proof coverage.

Arthur Freitas Ramos: Microsoft, USA; arfreita@microsoft.com
Ruy J. G. B. de Queiroz and Anjolina Grisi de Oliveira: Centro de Informática, Universidade
Federal de Pernambuco, Brazil;
ruy@cin.ufpe.br, ago@cin.ufpe.br
Tiago M. L. de Veras: Departamento de Matemática, Universidade Federal Rural de
Pernambuco, Brazil; tiago.veras@ufrpe.br
Corresponding author: Arthur Freitas Ramos

1. INTRODUCTION

Computational paths record equality as explicit finite traces of elementary transformations. Instead of retaining only the endpoints of an equality witness, they retain the sequence of steps by which one expression is transformed into another.

2020 *Mathematics Subject Classification.* Primary 22A22, 54B15; Secondary 03B38, 55Q05, 68V20, 54B30.

Key words and phrases. computational paths, rewriting, fundamental groupoid, quotient topology, topological groupoid.

The extra structure supports rewriting, normalization, and groupoid operations. It also poses a design problem for the semantics: a topological interpretation should be sound, yet it should not identify by definition every pair of traces with homotopic realizations. The trace calculus and its identity-type motivation are developed in [7, 30, 27].

We study this problem for continuous geometric data. A rewrite presentation specifies primitive geometric steps and named rewrites between parallel finite traces. Every named rewrite carries an endpoint-fixed homotopy between the corresponding realized paths. The induced rewrite equality is generated only by these named rules, congruence, and the structural groupoid laws. We call the relation *scoped* because the available identifications are controlled by the presentation rather than by all ambient homotopies.

Passing to rewrite classes raises a second, purely topological problem. Concatenation is continuous on the space of explicitly composable representatives, but the intended arrow space is a quotient. The ordinary set of composable quotient arrows is a pullback inside a product of quotient spaces, and products of quotient maps need not be quotient. So continuity of concatenation on representatives does not by itself give continuity of multiplication for the ordinary pullback topology. We handle this by first taking the quotient of explicitly composable representatives. Its quotient topology is the final composable-pair topology, for which multiplication is continuous by construction. A separate theorem compares it with the ordinary pullback topology.

The geometric paths are ordinary continuous maps $I \rightarrow X$. Where we require a homotopy to exist, a coherent representative does not record which homotopy was chosen. The coherence condition is proof-irrelevant in this sense, although the path itself is still data. A computational trace, on the other hand, records its primitive steps explicitly. The comparison with the fundamental groupoid reads these traces as geometric paths. It does not identify type-theoretic identity proofs with topological paths, and it assumes neither univalence nor higher-inductive types [17, 3, 15].

1.1. A computational path in one example. Suppose that a , b , and c are expressions and that r and s are elementary transformations

$$a \xrightarrow{r} b \xrightarrow{s} c.$$

The composite computational path $p = r; s$ records both steps. Its endpoints say that a is transformed into c , but the path also remembers the route through b . It has an identity, a reversal $p^{-1} = s^{-1}; r^{-1}$, and a composite with any compatible trace. A named rewrite may relate two parallel traces, for example

$$r; s \rightsquigarrow u, \quad a \xrightarrow{u} c,$$

but that identification is available only when the presentation declares it. A computational-path derivation is therefore a finite proof object, and its endpoints alone do not determine it.

In the geometric setting, each elementary step is realized by a continuous interval path. The word $r; s$ is realized by concatenating the realizations of r and s , and a named rewrite carries an endpoint-fixed homotopy between the two realizations. The basic semantic implication is

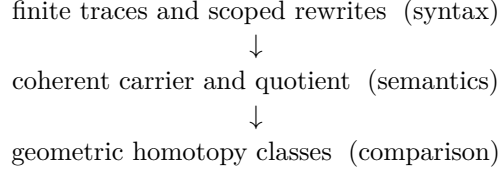
$$p \text{ has a scoped derivation to } q \implies |p| \simeq_{\text{rel } \partial I} |q|.$$

The syntax does not build in the converse. Whether the converse holds is the geometric-completeness problem of Section 6. For the one-generator circle, the same picture becomes the signed-word cancellation

$$a; a^{-1}; a \rightsquigarrow a,$$

a first glimpse of the normal-form argument in Section 8.

The construction has three layers:



The first transition equips traces with coherent geometric representatives and forms the scoped quotient. The second is the realization map to geometric homotopy classes. Soundness gives the forward comparison; normal forms or another completeness argument are needed to make it faithful. The notation used below follows this order: $\text{Tr}_{\mathcal{P}}$ is the flat word carrier, T is the common coherent carrier equipped with both a trace-sensitive topology T_{tr} and an observable topology T_{obs} , and $G_{\mathcal{P}}$ is the scoped quotient arrow space.

The main results address three questions.

- (i) *What information does the topology retain?* We define the trace-sensitive and observable topologies and prove that the identity from the former to the latter is continuous. Their scoped quotients can have different topologies on the same arrow set. A continuous trace section and geometric completeness make them agree. A complete presentation over the harmonic archipelago shows why the section hypothesis matters. Continuous substitutions of finite words for primitive steps preserve the trace-sensitive quotient when inverse substitutions are witnessed by scoped rewrites. Adding a redundant generator can nevertheless change the observable topology. Exact parametrization also matters: equal-slot and binary realization can give different observable quotients.
- (ii) *When is composition continuous?* We construct the groupoid with continuous multiplication on the final composable-pair domain. We give four equivalent conditions for this domain to have the ordinary pullback topology. Compact-Hausdorff, discreteness, and openness of the arrow quotient ensure agreement. The universal presentation gives a positive geometric case under local path-connectedness and semilocal simple connectivity; the Hawaiian-earring based-loop example shows that agreement can fail.
- (iii) *When do rewrites capture geometric equality?* We prove a completeness criterion using normal forms and identify the universal presentation with the quotient-topologized fundamental groupoid. The finite-generator circle and torus examples illustrate the criterion; a product theorem extends their based completeness to finite tori. For the two-circle rose, a genuine geometric covering detects reduced word order and proves completeness between represented loops. Maps of presentations also induce continuous maps of their quotient groupoids.

The quotient-space inputs and the circle and torus calculations are classical. For scoped rewrite presentations, we give the final-domain composition construction,

an exact criterion for ordinary-pullback composition, and a completeness interface for comparing rewrite classes with homotopy classes. The examples show that geometric completeness alone does not settle the quotient topology: a continuous trace section collapses the two topologies, while the complete harmonic-archipelago presentation separates them. Section 10 says which parts are checked in Lean, including the result registered in Palomar [32]. The presentation-change theorem and its redundant-generator counterexample give a further boundary: algebraically inverse interpretations preserve the full-word topology under explicit continuity conditions, while algebraic conservativity alone does not preserve the observable topology. The compact-open path-class openness and ordinary-groupoid theorems used in the positive case are due to Holkar, Hossain, and Kulkarni [18].

Our earlier work supplies the ingredients that this paper organizes into a topological semantics. The identity-type and explicit-calculus papers develop the interpretation of equality as a computational path and the associated rewrite operations [7, 30, 27]; the 2021 papers use those operations to calculate fundamental groups and fundamental groupoids [9, 26]. The weak-groupoid and weak ω -groupoid developments describe the higher coherence carried by the calculus [10, 28]. The topological application, the Lean formalization, and the Seifert–van Kampen preprint are geometric and formal case studies [8, 25, 29]. Here we give a scoped quotient for continuous presentations, distinguish the final and ordinary composable topologies, prove the universal property and the completeness interface, and recover the classical circle and torus calculations. The quotient records arrow classes while the earlier higher-coherence constructions retain witnesses between them. The book-length account of the calculus gives additional background [33]. The fundamental-groupoid terminology follows the standard topological treatment of groupoids and covering-space calculations [3, 15]. There is also a substantial literature on the topology of fundamental groupoids. Brown and Danesh-Narue construct a lifted topology making quotient groupoids topological groupoids under local hypotheses [4]; more recent Lasso and Brown topologies pursue the same goal by changing the topology on the fundamental-groupoid quotient [23, 24]. We keep the quotient topology on path classes, even though the quotient-topology literature shows that multiplication can fail to be jointly continuous already for the based fundamental group [12, 2]. Our final-domain construction locates that obstruction in the product-quotient comparison theorem, and Proposition 4.9 works it out for the Hawaiian earring.

The paper is organized as follows. Section 2 defines geometric traces, the two representative topologies, scoped presentations, and realization soundness. Section 3 constructs the quotient groupoid and proves its semantic factorization property. Section 4 compares the final and ordinary composable-pair topologies. Functoriality is treated in Section 5, and geometric comparison and completeness in Section 6. Section 7 gives the universal presentation; the circle and torus calculations are in Section 8. Section 9 discusses the logical and rewriting perspective, Section 10 describes what the Lean developments check, and Section 11 concludes.

2. GEOMETRIC TRACES AND SCOPED PRESENTATIONS

Throughout, X is a topological space and $I = [0, 1]$. Give the interval-path space X^I the compact-open topology. Let E be a topological space of oriented primitive steps, with continuous endpoint maps $s, t : E \rightarrow X$ and a continuous realization

map

$$\rho : E \longrightarrow X^I, \quad \rho(e)(0) = s(e), \quad \rho(e)(1) = t(e).$$

The continuity requirement is part of the presentation data. Let E^+ be a topological copy of E , let E^- be a formal inverse copy, and write $\tilde{E} = E^+ \sqcup E^-$. On E^+ we use the given endpoint and realization maps; on E^- the endpoint maps are interchanged and the realization is reversed. Thus there are continuous extensions

$$\tilde{s}, \tilde{t} : \tilde{E} \longrightarrow X, \quad \tilde{\rho} : \tilde{E} \longrightarrow X^I,$$

whose restrictions to E^+ are s, t, ρ and whose restrictions to the inverse copy (under $E^- \cong E$) are $t, s, e \mapsto \rho(e)^{-1}$, respectively.

Figure 1 separates the space of step labels from the space in which their paths run. The inverse copy does not require an orientation on the topological space E : only the source, target, and direction of the represented path are reversed.

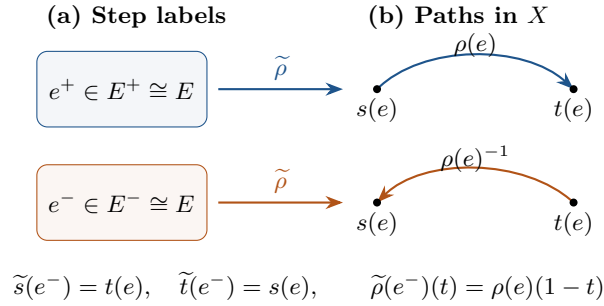


FIGURE 1. Formal reversal changes the direction of a represented path and leaves the topology of the step space unchanged. The two curves depict the same geometric route traversed in opposite directions. Endpoint maps take values in X ; the realization map takes values in X^I .

For composable interval paths α and β and $m, n \in \mathbb{N}$, write $\alpha \star_{m,n} \beta$ for the length-weighted concatenation. When $m, n > 0$, it is given by

$$(\alpha \star_{m,n} \beta)(t) = \begin{cases} \alpha\left(\frac{m+n}{m}t\right), & 0 \leq t \leq \frac{m}{m+n}, \\ \beta\left(\frac{(m+n)t - m}{n}\right), & \frac{m}{m+n} \leq t \leq 1. \end{cases}$$

Use the natural zero-length conventions $\alpha \star_{0,n} \beta = \beta$ and $\alpha \star_{m,0} \beta = \alpha$; when $m = n = 0$, use the constant path at the common endpoint. Thus the two input paths occupy m and n equal word slots, respectively. For composable paths α , β , and γ occupying m , n , and k word slots, respectively, this gives the following literal identities, including the zero-length cases under the stated conventions:

$$(\alpha \star_{m,n} \beta) \star_{m+n,k} \gamma = \alpha \star_{m,n+k} (\beta \star_{n,k} \gamma), \quad (\alpha \star_{m,n} \beta)^{-1} = \beta^{-1} \star_{n,m} \alpha^{-1}.$$

Example 2.1 (Reading the weights). Suppose that α occupies two word slots and β occupies one. Then $\alpha \star_{2,1} \beta$ follows α until time $2/3$ and follows β on the remaining third of the interval. If γ also occupies one slot, strict reassociation is visible in the concrete identity

$$(\alpha \star_{2,1} \beta) \star_{3,1} \gamma = \alpha \star_{2,2} (\beta \star_{1,1} \gamma).$$

The weights record word-slot lengths and nothing else: no choice of parenthesization and no geometric homotopy witness is involved.

Definition 2.2 (geometric trace). For $n \geq 1$, let $W_n \subseteq \tilde{E}^n$ be the subspace of composable words $(e_1, \dots, e_n)$ satisfying $\tilde{t}(e_i) = \tilde{s}(e_{i+1})$ for every i . Put $W_0 = X$, representing empty traces, with both endpoint maps equal to the identity on X , and define the topological trace carrier by the disjoint union

$$\mathrm{Tr}_{\mathcal{P}} = \coprod_{n \in \mathbb{N}} W_n.$$

Give $\mathbb{N}$ the discrete topology. The length map is the coproduct index and is continuous. For $p = (e_1, \dots, e_n) \in W_n$ with $n \geq 1$, define its realization by equal word slots,

$$|p|(t) = \tilde{\rho}(e_i)(nt - i + 1) \quad \text{for } 1 \leq i \leq n, \quad \frac{i-1}{n} \leq t \leq \frac{i}{n}.$$

For $p \in W_0 = X$, let $|p|$ be the constant path at its point. We write $\mathrm{Trace}(a, b)$ for the set of words with endpoints a, b . Empty words, word concatenation, and reversal are the identity, composition, and inverse operations on traces. In particular, for $p \in W_m$ and $q \in W_n$,

$$|p; q| = |p| \star_{m,n} |q|.$$

Representing traces as flat words makes the unit, associativity, double-inverse, and reversal-of-composite laws literal equalities. Their equal-slot realizations satisfy the same equalities. Inverse cancellation is different: a word followed by its reversal need not be the empty word, and its realization contracts by a homotopy.

Lemma 2.3 (continuity of trace realization). *The equal-slot realization map*

$$|\cdot| : \mathrm{Tr}_{\mathcal{P}} \longrightarrow X^I$$

is continuous for the coproduct trace topology.

Proof. On the zero-letter component $W_0 = X$, the realization is the constant-path map. Its adjoint $X \times I \rightarrow X$, $(a, t) \mapsto a$, is continuous, so the compact-open exponential law gives continuity into X^I . Fix $n \geq 1$. For each $1 \leq i \leq n$, the closed slot

$$C_i = W_n \times [(i-1)/n, i/n] \subseteq W_n \times I$$

supports the continuous formula

$$(p, t) \mapsto \tilde{\rho}(e_i(p))(nt - i + 1),$$

where $e_i : W_n \rightarrow \tilde{E}$ is the i th coordinate. Continuity follows from the continuity of $\tilde{\rho}$ and of evaluation for the compact-open topology. On a common boundary $t = i/n$, the two formulas agree because composability gives $\tilde{t}(e_i) = \tilde{s}(e_{i+1})$. The pasting lemma therefore gives a continuous adjoint $W_n \times I \rightarrow X$. Since I is compact Hausdorff, the compact-open exponential law gives a continuous map $W_n \rightarrow X^I$. The coproduct universal property now assembles these component maps into the displayed realization map. $\square$

The compact-open topology also gives continuous constant-path and reversal maps on X^I . On each fixed pair of length components, the weighted operation $\star_{m,n}$ is continuous on the corresponding endpoint pullback; the length coordinates are discrete, so these operations assemble continuously over the coproduct. Word concatenation and reversal are continuous on the corresponding endpoint pullbacks

of the W_n . Together with Lemma 2.3, these are the continuity facts used below. Nothing is assumed about products of quotient maps.

Let

$$\text{Coh}_{\mathcal{P}}(p, \gamma) : \Longleftrightarrow \text{there exists an endpoint-fixed homotopy from } \gamma \text{ to } |p|.$$

Only the existence of such a homotopy is required; the homotopy itself is not an entry of the pair. The coherent carrier is the set

$$T = \{(p, \gamma) \in \text{Tr}_{\mathcal{P}} \times X^I : \gamma(0) = s(p), \gamma(1) = t(p), \text{Coh}_{\mathcal{P}}(p, \gamma)\}.$$

The two entries have different jobs. The trace p retains the finite sequence of primitive steps to which the scoped rewrite relation applies, while γ is an independently chosen geometric representative observed by the topology. The coherence condition connects them without turning every ambient homotopy into a syntactic rewrite. The scoped quotient identifies all coherent representatives of the same scoped trace class. Thus γ affects the representative topologies and defines the raw realization map, but no particular choice of γ survives in a quotient arrow.

Geometric meaning of coherence. Having the same endpoints is necessary but not sufficient for coherence. Let $A = (-1, 0)$, $B = (0, 1)$, and $C = (1, 0)$ in $\mathbb{R}^2$. Let α traverse the two straight segments AB and BC at equal speed on the two halves of I , and put $\beta(t) = (2t - 1, 0)$. In the plane,

$$H(u, t) = (1 - u)\alpha(t) + u\beta(t)$$

is a continuous homotopy from α to β fixing A, C . Thus, if $|p| = \alpha$, the pair (p, β) is coherent. If instead $X = \mathbb{R}^2 \setminus \{(0, \frac{1}{3})\}$, both paths still exist, but the loop $\alpha \star_{1,1} \beta^{-1}$ surrounds the missing point once. Its winding is nonzero, so no endpoint-fixed homotopy exists in X [15]. Figure 2 depicts this change. Both paths remain continuous; what fails is that the loop between them cannot be filled inside the specified space.

More generally, for parallel paths α, β ,

$$\alpha \simeq_{\partial I} \beta \iff \alpha \star_{1,1} \beta^{-1} \simeq_{\partial I} c_{\alpha(0)}.$$

This follows from the groupoid laws of endpoint-fixed homotopy classes. A homotopy may be regarded as a path in $X^I(A, C)$, but the computational trace remains a finite word. A simple or discrete step space E does not remove obstructions in X either. Every named rewrite still needs a semantic homotopy, and the existence of a homotopy does not by itself declare a scoped rewrite.

We now put two canonical topologies on this same coherent carrier. First, let

$$\kappa_{\text{tr}} : T \longrightarrow \text{Tr}_{\mathcal{P}} \times X^I, \quad \kappa_{\text{tr}}(p, \gamma) = (p, \gamma).$$

The *trace-sensitive topology* T_{tr} is the initial topology induced by κ_{tr} ; because T is a subset of $\text{Tr}_{\mathcal{P}} \times X^I$, this is just the subspace topology. In it, the complete finite word is a topological coordinate. Second, let

$$\kappa_{\text{obs}} : T \longrightarrow X \times X \times \mathbb{N} \times X^I \times X^I, \quad \kappa_{\text{obs}}(p, \gamma) = (s(p), t(p), \ell(p), |p|, \gamma).$$

The *observable topology* T_{obs} is the initial topology induced by κ_{obs} : it is the coarsest topology making these semantic observables continuous. Since the endpoint maps, length, and realization are continuous on the trace carrier, κ_{obs} factors continuously through κ_{tr} . Consequently the identity

$$i_T : T_{\text{tr}} \longrightarrow T_{\text{obs}}$$

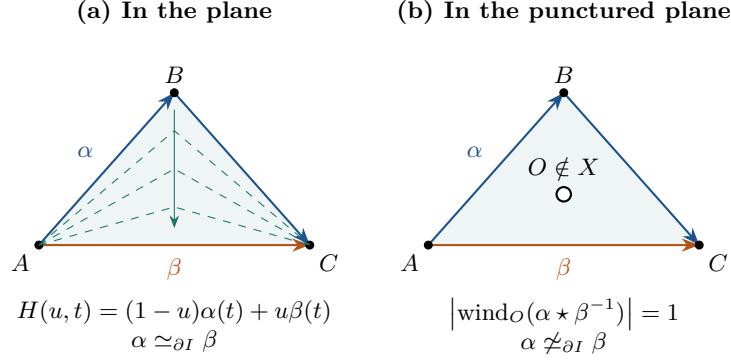


FIGURE 2. Coherence requires a homotopy inside the chosen ambient space. Left: intermediate broken paths move B to the segment AC , keeping A, C fixed. Right: deleting the interior point $O = (0, \frac{1}{3})$ prevents any such homotopy, although both boundary paths remain continuous. For $|p| = \alpha$, (p, β) is coherent only in the left-hand example. The shaded triangle is available in (a), but is not a filling in (b).

is continuous; that is, the trace-sensitive topology is finer than the observable one.

Unless a superscript says otherwise, Sections 3–8 use T_{obs} and its quotient, and we write $T := T_{\text{obs}}$ and $T^{(2)} := T_{\text{obs}}^{(2)}$. The results of Sections 3, 5, 6, and 7 hold verbatim for T_{tr} . Their proofs use continuity of raw maps on the carrier, which Lemma 2.10 and the coordinate arguments of Sections 5 and 7 give for both topologies, and quotient criteria that do not care which topology the carrier has. Geometric completeness is a property of equivalence relations, so it does not depend on the choice at all. Section 4 is the exception, since compatibility of the final and ordinary topologies has to be checked for each representative topology separately. Proposition 2.4 compares the two resulting quotients.

For the observable choice, write $\kappa = \kappa_{\text{obs}}$. Its five coordinates record the source, target, word length, realized trace, and chosen geometric path. They determine the topology, but not the equivalence relation used in the scoped quotient. Two distinct traces can have the same five coordinates without being related by the allowed rewrites.

For $* = \text{tr}, \text{obs}$, define the common composable set with its chosen subspace topology by

$$T_*^{(2)} = \{(u, v) \in T_* \times T_* : t(u) = s(v)\}.$$

Proposition 2.4 (trace-sensitive refinement). *Let T_{tr} and T_{obs} denote the two topologies above. The identity $i_T : T_{\text{tr}} \rightarrow T_{\text{obs}}$ is continuous, and its restriction to the common composable set is continuous:*

$$i_T^{(2)} : T_{\text{tr}}^{(2)} \longrightarrow T_{\text{obs}}^{(2)}.$$

Let $G_{\mathcal{P}}^{\text{tr}}$ and $G_{\mathcal{P}}^{\text{obs}}$ be the quotient arrow spaces for the same scoped relation, formed from the two representative topologies. Then the identity on quotient sets induces a continuous bijection

$$J : G_{\mathcal{P}}^{\text{tr}} \longrightarrow G_{\mathcal{P}}^{\text{obs}}, \quad J q_{\text{tr}} = q_{\text{obs}} i_T.$$

The corresponding map on final composable-pair quotients is continuous as well. In general J^{-1} need not be continuous, so J need not be a homeomorphism (Example 2.8).

Proof. The factorization of κ_{obs} through κ_{tr} proves continuity of i_T by the initial-topology criterion. Restriction to the endpoint pullback gives $i_T^{(2)}$. The identity preserves the scoped equivalence relation, so it descends to J . Since q_{tr} is a quotient map and $Jq_{\text{tr}} = q_{\text{obs}}i_T$ is continuous, the quotient criterion proves continuity of J . The same argument applies to the componentwise final composable-pair quotients. $\square$

Proposition 2.5 (when trace sensitivity disappears). *Let $P_{\mathcal{P}} \subseteq X^I$ be the space of paths represented by coherent pairs, with its subspace topology. Write $\pi_{\text{obs}} : T_{\text{obs}} \rightarrow P_{\mathcal{P}}$ and $\pi_{\text{tr}} : T_{\text{tr}} \rightarrow P_{\mathcal{P}}$ for the maps that forget the trace. Suppose that $\mathcal{P}$ is geometrically complete and that π_{tr} has a continuous section $s : P_{\mathcal{P}} \rightarrow T_{\text{tr}}$. Then the quotient comparison $J : G_{\mathcal{P}}^{\text{tr}} \rightarrow G_{\mathcal{P}}^{\text{obs}}$ is a homeomorphism.*

Proof. The observable code makes π_{obs} continuous. Thus $\theta = s\pi_{\text{obs}} : T_{\text{obs}} \rightarrow T_{\text{tr}}$ is continuous. For each coherent pair u , the chosen paths of u and $\theta(u)$ are equal. Geometric completeness makes their scoped classes equal, so $q_{\text{tr}}\theta = J^{-1}q_{\text{obs}}$. Since q_{obs} is quotient, J^{-1} is continuous. The forward map is continuous by Proposition 2.4. $\square$

The hypothesis is a continuous choice of trace for each represented path; completeness alone is not used to choose one. The universal one-letter section supplies such a choice. On the circle and torus based-loop presentations, a standard word selected by winding also gives a section: winding is locally constant in the compact-open based-loop space, and the geometric-path coordinate varies continuously. Thus the two quotient topologies agree on those based-loop examples.

Proposition 2.6 (completeness without trace-topology collapse). *There is a geometrically complete presentation for which the comparison $J : G_{\mathcal{P}}^{\text{tr}} \rightarrow G_{\mathcal{P}}^{\text{obs}}$ is not a homeomorphism. On the quotient formed from its fixed-endpoint coherent carrier at one base point, the trace-sensitive topology is discrete and the observable topology is indiscrete.*

Proof. Let X be the harmonic archipelago, with base point a . Take E to be the set X^I with the discrete topology, and let each step realize the path it labels. The endpoint and realization maps are continuous because E is discrete. Declare a rewrite between every pair of parallel traces whose realizations are endpoint-fixed homotopic. Soundness and the declared rules make this presentation geometrically complete.

In the fixed-endpoint trace carrier, each trace is an isolated point: this holds for positive-length words because E is discrete, and for the empty trace because its endpoint is fixed at a . For a scoped class C , its inverse image in $T_{\text{tr}}(a, a)$ is the union of the full coherent fibers over traces realizing C . Each such fiber is open in the trace-sensitive carrier. Thus every singleton is open in the based trace-sensitive quotient.

The projection $(p, \gamma) \mapsto \gamma$ from the observable based carrier to $X^I(a, a)$ is continuous. It has the continuous section $\gamma \mapsto (\langle \gamma \rangle, \gamma)$, whose observable code is $(a, a, 1, \gamma, \gamma)$. It is therefore a quotient map. Completeness identifies the scoped

relation with endpoint-fixed homotopy, so the based observable quotient is homeomorphic to $\pi_1^q(X, a)$. Fabel proved that this group is uncountable and indiscrete [11, Corollaries 9–10]; the uniform loop topology used there agrees with the compact-open topology for his metric model. The based comparison is a continuous bijection from a discrete space to a non-singleton indiscrete space.

The global comparison fails as well. Choose a nonidentity class C in $\pi_1^q(X, a)$. Its inverse image under the global trace-sensitive quotient contains no empty traces and is a union of full coherent fibers over positive-length words, hence is open. If $\{C\}$ were open in the global observable quotient, its pullback along the continuous one-letter section $X^I(a, a) \rightarrow T_{\text{obs}}$ would make the homotopy class C open in $X^I(a, a)$, contrary to the indiscrete quotient topology. $\square$

Example 2.7 (Same geometry, different computation). Suppose $p, q \in W_n$ have the same endpoints and the same equal-slot realization, and let γ be a common coherent representative. If $p \neq q$, then (p, γ) and (q, γ) have identical observable codes. They are therefore topologically indistinguishable in T_{obs} , even though the scoped quotient identifies them only when $p \simeq_{\mathcal{P}} q$. If the trace carrier is Hausdorff, the trace-sensitive topology T_{tr} can separate these two computationally different points. The allowed rewrites are the same in both cases; only the power of the topology to distinguish points changes.

Example 2.8 (A separation that survives the quotient). Let $X = \mathbb{R}/\mathbb{Z}$ and let $E = \{e, f\}$ be discrete. Both steps are based at 0 and realize the same nonconstant loop $\lambda(t) = [t]$. Declare no named rewrites; the scoped relation contains only the structural groupoid laws. Write

$$u_e = (e, \lambda), \quad u_f = (f, \lambda),$$

for two coherent points. Assign signed count $(1, 0)$ to e and $(0, 1)$ to f , extend by addition over concatenation, and negate under reversal. This map from traces to $\mathbb{Z}^2$ respects every structural groupoid law. It separates the scoped classes of u_e and u_f , despite their identical realized loops. Because the trace space retains the full word and E is discrete, the signed-count map is continuous to discrete $\mathbb{Z}^2$. It descends to a continuous map on $G_{\mathcal{P}}^{\text{tr}}$, whose inverse image of $\{(1, 0)\}$ is open, contains $q_{\text{tr}}(u_e)$, and excludes $q_{\text{tr}}(u_f)$.

The observable codes of u_e and u_f are identical: both have the same endpoints, length, realization, and coherent representative. They are therefore topologically indistinguishable in T_{obs} . An open subset of $G_{\mathcal{P}}^{\text{obs}}$ containing $q_{\text{obs}}(u_e)$ would pull back to an observable-open set containing u_e , hence also u_f . Thus it must contain $q_{\text{obs}}(u_f)$. The continuous bijection $J : G_{\mathcal{P}}^{\text{tr}} \rightarrow G_{\mathcal{P}}^{\text{obs}}$ from Proposition 2.4 is therefore not a homeomorphism. The difference comes from two labels for one nonconstant loop. Figure 3 summarizes the separation mechanism.

Example 2.9 (One trace, two coherent representatives). Let p be a trace and let γ and δ be endpoint-fixed homotopic representatives with

$$\gamma \simeq_{\partial I} |p| \simeq_{\partial I} \delta.$$

Then (p, γ) and (p, δ) are both points of T . The observable topology may observe the difference between their fifth coordinates, but the scoped quotient identifies them because the trace coordinate is the same. A coherent representative is therefore useful before quotienting without becoming extra data in a quotient arrow.

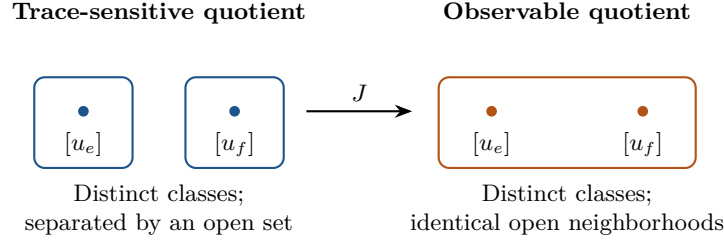


FIGURE 3. The two quotients identify the same arrows and differ only in their topology. In Example 2.8, both steps realize the same nonconstant circle loop, but their signed-count values differ. The map J preserves both scoped classes. In the observable quotient every open set containing either class contains both. Boxes illustrate separation properties only; the shared box is not asserted to be an open two-point subset.

For $u = (p, \gamma) \in T$, put $s(u) = s(p)$ and $t(u) = t(p)$.

Lemma 2.10 (continuity of coherent operations). *For either $*$ = tr or $*$ = obs, let $\iota : X \rightarrow T_*$ send a to the coherent identity $(1_a, c_a)$, let $\sigma : T_* \rightarrow T_*$ send (p, γ) to (p^{-1}, γ^{-1}) , and let*

$$\mu_* : T_*^{(2)} \longrightarrow T_*, \quad ((p, \gamma), (q, \delta)) \longmapsto (p; q, \gamma \star_{\ell(p), \ell(q)} \delta),$$

where the weighted operation uses the discrete trace lengths. Then all three maps are continuous.

Proof. For T_{obs} , it is enough to check continuity after applying κ_{obs} . The three resulting observable codes are respectively

$$(a, a, 0, c_a, c_a), \quad (t(p), s(p), \ell(p), |p|^{-1}, \gamma^{-1}),$$

and, on the composable pullback,

$$(s(p), t(q), \ell(p) + \ell(q), |p| \star_{\ell(p), \ell(q)} |q|, \gamma \star_{\ell(p), \ell(q)} \delta).$$

The first is continuous because constant paths depend continuously on a ; the second uses reversal on X^I ; and the third uses addition on the discrete length coordinate and continuity of the weighted operation on each fixed length component. Each code depends continuously only on the observable codes of the input(s), so the initial-topology criterion proves continuity of all three operations. For T_{tr} , the corresponding first coordinates are the continuous word maps $a \mapsto 1_a$, $p \mapsto p^{-1}$, and $(p, q) \mapsto p; q$ on the trace carrier; the second coordinates are the same continuous path operations just used. The initial-topology criterion for κ_{tr} therefore proves continuity in the trace-sensitive topology as well. Reversal and weighted concatenation preserve the coherence predicate by reversing and concatenating the endpoint-fixed homotopies. The Lean development flattens the parenthesized internal trace to a signed word so that these first-coordinate maps can be checked. $\square$

Definition 2.11 (scoped presentation). A continuous geometric rewrite presentation $\mathcal{P}$ on (X, E) consists of a predicate

$$\text{Rule}_{\mathcal{P}}(p, q), \quad p, q : \text{Trace}(a, b),$$

and, for every instance of $\text{Rule}_{\mathcal{P}}(p, q)$, an endpoint-fixed homotopy between the realizations of p and q .

Here $\text{Rule}_{\mathcal{P}}$ specifies the named rewrites; it should not be confused with the realization comparison $R_{\mathcal{P}}$ of Section 6. The definition asks for a homotopy witness for each named rewrite and imposes no completeness condition.

Definition 2.12 (scoped rewrite equality). For parallel traces, write $p \simeq_{\mathcal{P}} q$ for the inductively generated relation whose constructors are reflexivity, the named generators $\text{Rule}_{\mathcal{P}}$, symmetry, transitivity, congruence under concatenation and reversal, and the following structural groupoid coherences:

$$\begin{array}{lll} (1_a); p \simeq_{\mathcal{P}} p, & p; (1_b) \simeq_{\mathcal{P}} p, & (p; q); r \simeq_{\mathcal{P}} p; (q; r), \\ p^{-1}; p \simeq_{\mathcal{P}} 1_b, & p; p^{-1} \simeq_{\mathcal{P}} 1_a, & (p^{-1})^{-1} \simeq_{\mathcal{P}} p, \\ (1_a)^{-1} \simeq_{\mathcal{P}} 1_a, & (p; q)^{-1} \simeq_{\mathcal{P}} q^{-1}; p^{-1}. \end{array}$$

Only the rules declared by $\mathcal{P}$ and these structural constructors are available. Literal equality of traces is handled by reflexivity and substitution; ambient homotopy of their realizations is not an additional constructor.

Lemma 2.13 (weighted concatenation invariance). *Let $\alpha \simeq_{\partial I} \alpha'$ and $\beta \simeq_{\partial I} \beta'$ be endpoint-fixed homotopies of composable paths. Write c_x for the constant path at x , and assume explicitly that*

$$\begin{aligned} m = 0 &\Rightarrow \alpha(0) = \alpha(1) \text{ and } \alpha \simeq_{\partial I} c_{\alpha(0)}, \\ n = 0 &\Rightarrow \beta(0) = \beta(1) \text{ and } \beta \simeq_{\partial I} c_{\beta(0)}, \\ m' = 0 &\Rightarrow \alpha'(0) = \alpha'(1) \text{ and } \alpha' \simeq_{\partial I} c_{\alpha'(0)}, \\ n' = 0 &\Rightarrow \beta'(0) = \beta'(1) \text{ and } \beta' \simeq_{\partial I} c_{\beta'(0)}. \end{aligned}$$

For arbitrary $m, n, m', n' \in \mathbb{N}$, with the zero-length conventions above,

$$\alpha \star_{m,n} \beta \simeq_{\partial I} \alpha' \star_{m',n'} \beta'.$$

Proof. When all four weights are positive, let $\lambda_{m,n} : [0, 1] \rightarrow [0, 1]$ be the piecewise-linear endpoint-fixing homeomorphism that maps $m/(m+n)$ to $1/2$ and is linear on the two sides. Explicitly,

$$\lambda_{m,n}(t) = \begin{cases} \frac{m+n}{2m}t, & 0 \leq t \leq \frac{m}{m+n}, \\ \frac{1}{2} + \frac{m+n}{2n} \left(t - \frac{m}{m+n} \right), & \frac{m}{m+n} \leq t \leq 1. \end{cases}$$

Then

$$\alpha \star_{m,n} \beta = (\alpha \star_{1,1} \beta) \circ \lambda_{m,n}.$$

The straight-line homotopy $(1-u)\lambda_{m,n} + u\lambda_{m',n'}$ fixes 0 and 1, so changing the breakpoint gives an endpoint-fixed reparametrization homotopy. Horizontal concatenation of the given homotopies at the common breakpoint $1/2$ gives

$$\alpha \star_{1,1} \beta \simeq_{\partial I} \alpha' \star_{1,1} \beta'.$$

Combining these homotopies proves the claim when all weights are positive. If a weight is zero, first use the corresponding zero-slot hypothesis to replace the corresponding factor by the constant path at its endpoint. Such a factor has coinciding endpoints by the displayed hypothesis, so the stated convention then deletes that constant factor. When the zero/nonzero status changes between the two sides, this

replacement inserts or removes the factor up to endpoint-fixed homotopy. The remaining positive-weight factors are handled by the preceding reparametrization argument; when both weights are zero, the convention yields the common endpoint constant. Thus all zero-length cases follow as well. $\square$

Theorem 2.14 (realization soundness). *If $p \simeq_{\mathcal{P}} q$, then the realized paths are endpoint-fixed homotopic.*

Proof. Induct on the derivation of $p \simeq_{\mathcal{P}} q$. Reflexivity is the constant homotopy, and a named generator is sound by the corresponding presentation witness. Symmetry and transitivity use symmetry and vertical composition of endpoint-fixed homotopies. In the concatenation-congruence case, apply the weighted concatenation invariance lemma to the two inductive homotopies and the four trace lengths; this accounts for rewrites that change word length. The zero-weight hypotheses of that lemma hold because a trace of length zero is an empty word, whose realization is a constant path. For reversal, reverse the path parameter, replacing a homotopy $H(s, t)$ by $H(s, 1 - t)$. Unit, associativity, double-reversal, and reversal of a composite are literal equalities for the flat word realization, hence use reflexive homotopies. The two inverse-cancellation cases use the standard endpoint-fixed contraction homotopies made explicit below. This covers every rule that can occur in a scoped derivation. $\square$

Lemma 2.15 (backtracking contracts in every space). *For any continuous path $\alpha : I \rightarrow X$, the formula*

$$H(u, t) = \alpha(\min\{2t, 2(1 - t), 1 - u\}) \quad (u, t \in I)$$

is an endpoint-fixed homotopy from $\alpha \star_{1,1} \alpha^{-1}$ to $c_{\alpha(0)}$.

Proof. The minimum is a continuous map $I^2 \rightarrow I$, so H is continuous. At $u = 0$ its argument is the triangular function $\min\{2t, 2(1 - t)\}$, which traverses α and then retraces it. At $u = 1$ it is identically zero, and for every u its values at $t = 0, 1$ are zero. These are the required boundary conditions. Applying the same construction to α^{-1} proves the other inverse law. $\square$

The homotopy stays entirely in $\alpha(I)$: no disk in X is needed. For a nonempty trace p , the two factors in $p; p^{-1}$ have equal length, so its weighted realization is exactly the half-interval concatenation in Lemma 2.15; the empty case is constant. Figure 4 makes the contrast with Figure 2 explicit.

Remark 2.16. The soundness theorem is one-way. Equality of realized geometric homotopy classes does not imply scoped rewrite equality unless a separate geometric completeness theorem is proved. Because the implication runs in only one direction, the presentation can retain computational information.

3. THE QUOTIENT GROUPOID AND ITS UNIVERSAL PROPERTY

For coherent traces $u = (p, \gamma)$ and $v = (q, \delta)$, write $u \approx v$ when their endpoints agree and $p \simeq_{\mathcal{P}} q$. By realization soundness, equivalent coherent traces determine the same endpoint-fixed homotopy class of geometric paths. The quotient arrow space is

$$G_{\mathcal{P}} = T / \approx,$$

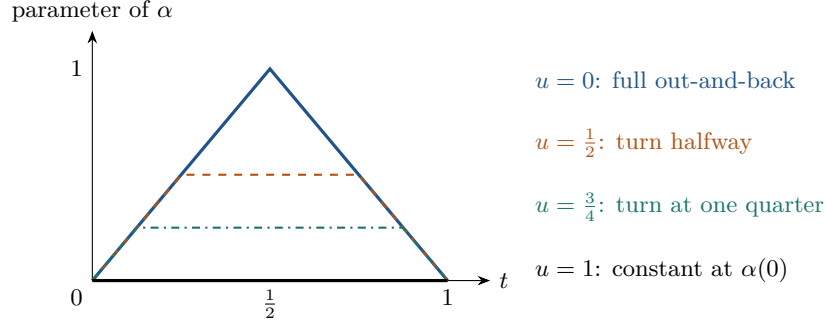


FIGURE 4. Backtracking cancellation as an explicit homotopy. The plotted function is $r_u(t) = \min\{2t, 2(1-t), 1-u\}$; the actual path is $H(u, t) = \alpha(r_u(t))$. As u increases, the turning point retreats along α and the path waits there before returning. All intermediate paths stay in $\alpha(I)$, so even a hole surrounded by α cannot obstruct this cancellation.

with the quotient topology induced by $q : T \rightarrow G_{\mathcal{P}}$. Endpoint maps descend to continuous maps

$$s, t : G_{\mathcal{P}} \longrightarrow X.$$

Reflexive paths and reversal descend because the scoped relation is closed under the corresponding constructors.

The explicit composable carrier is the $T^{(2)}$ defined in Section 2. Its quotient relation is componentwise scoped equivalence, and its quotient is denoted $G_{\mathcal{P}}^{(2), \text{fin}}$. Write

$$q_{\text{fin}}^{(2)} : T^{(2)} \longrightarrow G_{\mathcal{P}}^{(2), \text{fin}}$$

for the quotient map. Concatenation of representatives gives a well-defined map

$$m_{\text{fin}} : G_{\mathcal{P}}^{(2), \text{fin}} \longrightarrow G_{\mathcal{P}}.$$

The composite $q \circ \mu : T^{(2)} \rightarrow G_{\mathcal{P}}$ is continuous and constant on equivalent pairs. The quotient criterion therefore makes m_{fin} continuous. This argument uses the quotient of composable representatives, not a product of quotient maps.

Definition 3.1 (final-domain topological groupoid). Let $G \rightrightarrows X$ be a groupoid with topologies on G and X such that source, target, identity, and inverse are continuous. Give the set of composable arrow pairs $G \times_X G$ a topology, denoted C_{fin} , for which both projections $C_{\text{fin}} \rightarrow G$ are continuous. We call these data a *final-domain topological groupoid* when multiplication

$$m : C_{\text{fin}} \longrightarrow G$$

is continuous. The groupoid laws are the usual algebraic ones. The topology C_{fin} is part of the structure. It is at least as fine as the subspace topology from $G \times G$, and it may be strictly finer.

Taken alone, the definition is weak: the discrete topology on C_{fin} always qualifies. What matters below is that $G_{\mathcal{P}}$ comes with a canonical choice, the quotient topology from explicitly composable representatives. Section 4 compares that choice with the subspace topology.

Theorem 3.2 (canonical final-domain groupoid). *For every continuous geometric rewrite presentation $\mathcal{P}$:*

- (a) *s, t , the identity map $X \rightarrow G_{\mathcal{P}}$, and reversal $G_{\mathcal{P}} \rightarrow G_{\mathcal{P}}$ are continuous;*
- (b) *m_{fin} and the two projections $G_{\mathcal{P}}^{(2), \text{fin}} \rightarrow G_{\mathcal{P}}$ are continuous;*
- (c) *the descended operations satisfy the left and right unit laws, inverse laws, and associativity.*

Thus $G_{\mathcal{P}}$ is a final-domain topological groupoid.

Proof. By Lemma 2.10, the raw identity, reversal, and concatenation maps are continuous for the stated topology. Continuity of the endpoint maps and identities follows from the quotient criterion, because the corresponding raw maps are continuous. The same argument applies to reversal. The raw concatenation map $T^{(2)} \rightarrow T$ is continuous and respects the componentwise quotient relation, so it descends to m_{fin} ; continuity follows from the defining quotient topology on $G_{\mathcal{P}}^{(2), \text{fin}}$. The i th projection satisfies $\text{pr}_i \circ q_{\text{fin}}^{(2)} = q \circ \text{pr}_i$, so it is continuous by the same criterion.

For the algebraic laws, choose representatives. In the flat word model, left and right units, associativity, double reversal, and reversal of a composite are literal equalities of traces and of the weighted realizations; their scoped witnesses are therefore reflexive structural coherences. The two inverse laws are not literal equalities. They need endpoint-fixed contraction homotopies and are represented by the constructors $p^{-1}; p \simeq_{\mathcal{P}} 1_b$ and $p; p^{-1} \simeq_{\mathcal{P}} 1_a$. The quotient soundness criterion then turns each scoped derivation into equality of quotient arrows. The result is independent of representative choice because concatenation was defined through the componentwise quotient. $\square$

The next theorem describes which interpretations of coherent traces pass through $G_{\mathcal{P}}$. Such an interpretation must respect the rewrites and the groupoid operations. It must also give the same value to (p, γ) and (p, δ) whenever both are coherent: the scoped quotient does not retain the choice of geometric representative.

Theorem 3.3 (semantic factorization). *Let $\mathcal{H} \rightrightarrows X$ be a topological groupoid, and let $\varphi : T \rightarrow \mathcal{H}_1$ be continuous. Assume that φ preserves endpoints, identities, reversal, and concatenation, and that every named rewrite of $\mathcal{P}$ is sent to an equality in $\mathcal{H}_1$ for every choice of coherent representatives. Assume moreover that φ is invariant under changing the coherent representative: whenever $(p, \gamma), (p, \delta) \in T$, one has*

$$\varphi(p, \gamma) = \varphi(p, \delta).$$

Then there is a unique continuous map

$$\bar{\varphi} : G_{\mathcal{P}} \longrightarrow \mathcal{H}_1$$

such that $\bar{\varphi}q = \varphi$. The map $\bar{\varphi}$ preserves all groupoid operations, with composition on the source interpreted on $G_{\mathcal{P}}^{(2), \text{fin}}$.

Proof. Every trace p has the coherent representative $(p, |p|)$, and representative invariance gives $\varphi(p, \gamma) = \varphi(p, |p|)$ for every coherent (p, γ) . It therefore suffices to show that $p \simeq_{\mathcal{P}} q$ implies $\varphi(p, |p|) = \varphi(q, |q|)$, by induction on the derivation. Named rewrites are covered by hypothesis, and reflexivity, symmetry, and transitivity are immediate. Since $\mu((p, |p|), (q, |q|)) = (p; q, |p; q|)$ and $\sigma(p, |p|) = (p^{-1}, |p^{-1}|)$, preservation of concatenation and reversal handles the two congruence cases. The

structural coherences hold because $\mathcal{H}$ satisfies the groupoid laws and φ preserves identities, reversal, and concatenation. Hence φ is constant on $\approx$. The quotient universal property gives a unique map $\bar{\varphi}$, and the quotient criterion gives continuity. The operation-preservation equations hold on representatives and hence descend. For composition, the same argument is applied to the quotient map $T^{(2)} \rightarrow G_{\mathcal{P}}^{(2),\text{fin}}$; no product-quotient assumption is needed. $\square$

4. FINAL AND ORDINARY COMPOSABLE-PAIR TOPOLOGIES

As before, $G_{\mathcal{P}} = G_{\mathcal{P}}^{\text{obs}}$ and $T = T_{\text{obs}}$. The definitions and theorems of this section apply equally to the trace-sensitive quotient. Compatibility for one representative topology does not imply compatibility for the other, however; the continuous map J of Proposition 2.4 does not transfer it in either direction.

There is another natural composable domain. Let

$$G_{\mathcal{P}}^{(2),\text{ord}} = \{(x, y) \in G_{\mathcal{P}} \times G_{\mathcal{P}} : t(x) = s(y)\}$$

with the subspace topology. The underlying set of this space is the same as the underlying set of the final domain, but its topology need not be the same. The canonical ordinary-pair map is

$$q_{\text{ord}}^{(2)} : T^{(2)} \longrightarrow G_{\mathcal{P}}^{(2),\text{ord}}.$$

It is continuous and surjective. There is a unique bijection

$$j : G_{\mathcal{P}}^{(2),\text{fin}} \longrightarrow G_{\mathcal{P}}^{(2),\text{ord}}$$

such that

$$q_{\text{ord}}^{(2)} = j \circ q_{\text{fin}}^{(2)}.$$

The map j is continuous because $q_{\text{fin}}^{(2)}$ is quotient. Products of quotient maps need not be quotient maps, so $q_{\text{ord}}^{(2)}$ being quotient is an extra property; it does not follow formally from q being quotient.

Definition 4.1 (product-quotient compatibility). The presentation has product-quotient compatibility if $q_{\text{ord}}^{(2)}$ is a quotient map onto the ordinary composable-pair space. Since $q_{\text{ord}}^{(2)} = j \circ q_{\text{fin}}^{(2)}$ and $q_{\text{fin}}^{(2)}$ is quotient, this is equivalent to j being a quotient map.

The two domains have the same elements: composable pairs of scoped classes. They differ only in their open sets. For the final topology, a set U is open exactly when

$$(q_{\text{fin}}^{(2)})^{-1}(U) \text{ is open in } T^{(2)}.$$

For the ordinary topology, openness is inherited from $G_{\mathcal{P}} \times G_{\mathcal{P}}$. The final topology is always at least as fine as the ordinary one, since j is continuous. Product-quotient compatibility says that there are no additional open sets in the final topology.

Theorem 4.2 (ordinary/final comparison). *For every presentation $\mathcal{P}$, the following are equivalent:*

- (1) $q_{\text{ord}}^{(2)} : T^{(2)} \rightarrow G_{\mathcal{P}}^{(2),\text{ord}}$ is a quotient map;
- (2) $j : G_{\mathcal{P}}^{(2),\text{fin}} \rightarrow G_{\mathcal{P}}^{(2),\text{ord}}$ is a quotient map;
- (3) the inverse bijection $j^{-1} : G_{\mathcal{P}}^{(2),\text{ord}} \rightarrow G_{\mathcal{P}}^{(2),\text{fin}}$ is continuous;

(4) j is a homeomorphism, equivalently the final and ordinary composable-pair topologies coincide.

Thus each condition is also equivalent to product-quotient compatibility. When these conditions hold, multiplication is continuous from the ordinary pullback domain. Theorem 3.2 uses none of these conditions.

Proof. If j is quotient, so is the composite $q_{\text{ord}}^{(2)} = j \circ q_{\text{fin}}^{(2)}$. Conversely, suppose $q_{\text{ord}}^{(2)}$ is quotient and $j^{-1}(U)$ is open. Then $(q_{\text{ord}}^{(2)})^{-1}(U) = (q_{\text{fin}}^{(2)})^{-1}(j^{-1}(U))$ is open, so U is open. This proves that (1) and (2) are equivalent.

The map j is a continuous bijection. A bijective quotient map has continuous inverse, so (2) and (3) are equivalent; together with the already known continuity of j , these are equivalent to (4). The homeomorphism condition is precisely equality of the two topologies on the common underlying set. Finally, when these conditions hold, multiplication is continuous by the quotient criterion applied to its continuous representative-level concatenation. $\square$

Remark 4.3 (terminology). In the standard definition of a topological groupoid, multiplication is continuous on the ordinary pullback $G_{\mathcal{P}} \times_X G_{\mathcal{P}}$. Accordingly, the phrase “topological groupoid” without qualification is reserved for the case in which product-quotient compatibility has been proved. The unconditional object of Theorem 3.2 is instead called a *final-domain topological groupoid*: it has the same arrow set and groupoid laws, but its multiplication domain carries the final topology from explicitly composable representatives.

Remark 4.4 (comparison with convenient categories). One can instead change the ambient category of spaces, for example by using compactly generated spaces and their k -ified products [20]. That remedy changes the topology assigned to the ordinary product. The present construction stays in **Top** and records the final topology forced by the chosen representative quotient, so the failure of the ordinary product remains visible. We do not claim that the two remedies agree in general.

$$\begin{array}{ccccc}
 T^{(2)} & \xrightarrow{q_{\text{fin}}^{(2)}} & G_{\mathcal{P}}^{(2), \text{fin}} & \xrightarrow{m_{\text{fin}}} & G_{\mathcal{P}} \\
 \downarrow q_{\text{ord}}^{(2)} & & \downarrow j & & \parallel \\
 G_{\mathcal{P}}^{(2), \text{ord}} & \xlongequal{\quad} & G_{\mathcal{P}}^{(2), \text{ord}} & \xrightarrow{m_{\text{ord}}} & G_{\mathcal{P}}
 \end{array}$$

FIGURE 5. The composition-continuity boundary. The upper multiplication is always continuous. The comparison j is always a continuous bijection, and is a homeomorphism exactly under Theorem 4.2’s equivalent quotient conditions. These conditions imply continuity of m_{ord} . The paper does not prove the converse, that continuity of m_{ord} alone implies them.

Proposition 4.5 (open-map criterion). *If the bijection from the final composable quotient to the ordinary pair space is an open map, then product-quotient compatibility holds. In particular, an ordinary-pair map $q_{\text{ord}}^{(2)}$ that is open is quotient, so multiplication is continuous for the ordinary pullback topology.*

Proof. An open continuous bijection is a homeomorphism, so Theorem 4.2 applies. An open continuous surjection is a quotient map. $\square$

Theorem 4.6 (open arrow quotients). *If the arrow quotient $q : T \rightarrow G_{\mathcal{P}}$ is open, then $q_{\text{ord}}^{(2)} : T^{(2)} \rightarrow G_{\mathcal{P}}^{(2),\text{ord}}$ is an open quotient map. Consequently the final and ordinary composable-pair topologies agree, and multiplication is continuous on the ordinary pullback.*

Proof. The product $q \times q : T \times T \rightarrow G_{\mathcal{P}} \times G_{\mathcal{P}}$ is an open surjection. Because scoped equivalence preserves endpoints, $T^{(2)}$ is the full inverse image of $G_{\mathcal{P}}^{(2),\text{ord}}$ under this product map. Restricting an open map to the full inverse image of a subspace gives an open map onto that subspace: the image of $U \cap T^{(2)}$, for U open in $T \times T$, is $(q \times q)(U) \cap G_{\mathcal{P}}^{(2),\text{ord}}$. This restriction is $q_{\text{ord}}^{(2)}$, and it is continuous and surjective. It is therefore quotient, so Theorem 4.2 applies. $\square$

Theorem 4.7 (compact-Hausdorff compatibility). *Suppose that the final composable domain $G_{\mathcal{P}}^{(2),\text{fin}}$ is compact and that the ordinary composable domain $G_{\mathcal{P}}^{(2),\text{ord}}$ is Hausdorff. Then product-quotient compatibility holds. Hence the final and ordinary composable-pair topologies agree and ordinary-pullback multiplication is continuous.*

Proof. The comparison map

$$j : G_{\mathcal{P}}^{(2),\text{fin}} \longrightarrow G_{\mathcal{P}}^{(2),\text{ord}}$$

is a continuous bijection. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism: its image of every closed set is compact, hence closed, so its inverse is continuous. The composite $j \circ q_{\text{fin}}^{(2)} = q_{\text{ord}}^{(2)}$ is therefore a quotient map. This is exactly product-quotient compatibility. $\square$

Corollary 4.8 (discrete arrow spaces). *If $G_{\mathcal{P}}$ is discrete, then the final and ordinary composable-pair topologies coincide, and both are discrete.*

Proof. The ordinary domain is a subspace of the discrete space $G_{\mathcal{P}} \times G_{\mathcal{P}}$, so it is discrete. Every subset of the final domain is then the preimage under the continuous map j of an open set, hence open. So both topologies are discrete and j is a homeomorphism. $\square$

In both results, Theorem 4.2 turns agreement of the two topologies into continuity of ordinary multiplication. Theorem 3.2 needs neither hypothesis.

4.1. A strict failure: the Hawaiian earring. Let

$$\mathbb{H} = \bigcup_{n \geq 1} \left\{ (x, y) \in \mathbb{R}^2 : \left(x - \frac{1}{n} \right)^2 + y^2 = \frac{1}{n^2} \right\}$$

be the Hawaiian earring, based at $x_0 = (0, 0)$. Write $\Omega = \Omega(\mathbb{H}, x_0)$ for its based loop space with the compact-open topology and

$$q_{\Omega} : \Omega \longrightarrow \pi_1^q(\mathbb{H}, x_0)$$

for the endpoint-fixed homotopy quotient. Fabel proved that multiplication in $\pi_1^q(\mathbb{H}, x_0)$ is not continuous [12]. Loop concatenation $\Omega \times \Omega \rightarrow \Omega$ is continuous and induces this multiplication, so $q_{\Omega} \times q_{\Omega}$ is not a quotient map either.

Proposition 4.9 (Hawaiian-earring obstruction). *Use the observable based fiber of the universal presentation and the fiberwise quotient construction of Lemma 7.2:*

$$T_0 := T_{\mathcal{U}_{\text{H}}}^{\text{obs}}(x_0, x_0), \quad G_0 := G_{\mathcal{U}_{\text{H}}}^{\text{obs}}(x_0, x_0),$$

with quotient map $q_0 : T_0 \rightarrow G_0$ and projection $\pi_0 : T_0 \rightarrow \Omega$. Then the ordinary pair map

$$q_0 \times q_0 : T_0 \times T_0 \longrightarrow G_0 \times G_0$$

is not quotient. Consequently the final topology on composable pairs is strictly finer than the ordinary product topology, and multiplication on the ordinary product is discontinuous.

Proof. The fiberwise lemma gives that π_0 is quotient and has the continuous section $\sigma(\gamma) = (\langle \gamma \rangle, \gamma)$. It also gives a fiberwise universal-completion homeomorphism

$$h : G_0 \longrightarrow \pi_1^q(\mathbb{H}, x_0), \quad h \circ q_0 = q_\Omega \circ \pi_0.$$

Suppose that $q_0 \times q_0$ were quotient. Postcomposition with the homeomorphism $h \times h$ would make

$$(q_\Omega \times q_\Omega) \circ (\pi_0 \times \pi_0)$$

a quotient map. If $(q_\Omega \times q_\Omega)^{-1}(U)$ were open, its preimage under the continuous map $\pi_0 \times \pi_0$ would therefore be open; the quotient criterion for the displayed composite would force U to be open. Hence $q_\Omega \times q_\Omega$ would be quotient, contradicting Fabel's theorem.

Theorem 4.2 and its proof apply to the based fiber without change, with $T_0 \times T_0$ in place of $T^{(2)}$. They show that the continuous bijection from the final pair domain to the ordinary one is not a homeomorphism. Since the final topology is always at least as fine, it is strictly finer here. Finally, h preserves loop multiplication, so continuity of ordinary-product multiplication on G_0 would imply continuity on $\pi_1^q(\mathbb{H}, x_0)$, again contradicting Fabel's theorem. $\square$

Proposition 4.9 is stated for the observable quotient used in Theorem 4.2. For this universal presentation, Corollary 7.4 also applies on the based carrier: the one-letter path section is continuous for the trace-sensitive topology and recovers the same homotopy class. Its product with itself is a continuous section on based composable representatives. Repeating the product-quotient argument above with that section transfers the strict final/ordinary pair distinction and the discontinuity of ordinary multiplication to the trace-sensitive based quotient. The general refinement map alone would not establish that conclusion. The pair-section argument here is mathematical; the formal selection of the universal-collapse module checks the arrow comparison.

The discontinuity result is Fabel's. We use it to show that the two composable-pair topologies do differ, and that the failure sits at the inverse of the comparison map j . With the final topology, multiplication is continuous here as it is for every presentation, while for the ordinary product topology it is not.

5. FUNCTORIALITY

Let $\mathcal{P}$ and $\mathcal{Q}$ be presentations on (X, E) and (Y, F) . A presentation map consists of a continuous map $f : X \rightarrow Y$, a continuous step map preserving endpoints and realizations, and a proof that every named rule of $\mathcal{P}$ maps to a scoped derivation in $\mathcal{Q}$. Presentation maps form a category: the identity has the identity space and

step maps, and composition is the composite of the corresponding space and step maps, with the scoped-derivation proofs composed by induction.

Theorem 5.1 (functoriality). *Every presentation map $F : \mathcal{P} \rightarrow \mathcal{Q}$ induces a continuous map*

$$F_1 : G_{\mathcal{P}} \longrightarrow G_{\mathcal{Q}}$$

that preserves source, target, identities, reversal, and final-domain composition. The assignment preserves identity maps and composites.

Proof. Write $f : X \rightarrow Y$ for the underlying map of spaces and F_*p for the trace obtained by applying the step map to every letter of p . The raw map is

$$\tilde{F} : T_{\mathcal{P}} \longrightarrow T_{\mathcal{Q}}, \quad (p, \gamma) \longmapsto (F_*p, f \circ \gamma).$$

It preserves coherence because realizations are preserved by the step map and endpoint-fixed homotopies may be postcomposed with f . It is continuous: on every word stratum the induced trace map is continuous, postcomposition $X^I \rightarrow Y^I$ is continuous for the compact-open topology, and the observable code of $\tilde{F}(p, \gamma)$ is consequently a continuous function of the observable code of (p, γ) . Induction on scoped rewrite derivations proves that $\tilde{F}$ sends related traces to related traces. It therefore descends through the quotient, and the quotient criterion gives continuity on arrows. Applying the step map to reflexivity, reversal, and concatenation gives the corresponding preservation equations before quotienting. The same argument on the explicit composable carrier gives a continuous map on final composable domains and proves final composition preservation. Identity and composite step maps are definitionally the identity and composite maps on raw traces; quotient extensionality completes the proof. $\square$

The same construction gives ordinary-pair continuity for every presentation map. If both source and target presentations satisfy product-quotient compatibility, the ordinary multiplication maps are continuous as well. Continuity of the induced map on arrows does not require this additional compatibility assumption. For the trace-sensitive topology, the word component of $\tilde{F}$ is continuous on each coproduct stratum and the path component is the same postcomposition map, so the proof carries over.

5.1. Finite-word interpretations and a conservative abbreviation. A primitive may instead be interpreted as a finite trace with the same endpoints. Extend such a substitution M through identities, composition, and reversal. If images of every named source rule are target scoped rewrites, induction proves preservation of the entire scoped relation. Primitive roundtrip rewrites for M, N then extend to every trace. This is the algebraic part of a presentation change. It does not by itself establish continuity when length and exact realization are observable.

The fixed-endpoint criterion and the observable counterexample use endpoint-indexed *tree* trace carriers $\mathcal{T}_{\mathcal{P}}(a, b)$ and their fixed-endpoint scoped quotients $\mathcal{Q}_{\mathcal{P}}(a, b)$, with the stated trace topologies. Their Lean claims therefore do not assume the literal-word correspondence discussed in Section 10.

Proposition 5.2 (fixed-endpoint substitution criterion). *Let M, N be endpoint-preserving finite-trace substitutions in opposite directions, preserving named rules. Suppose their primitive roundtrips are scoped rewrites to the corresponding primitive and their recursively extended trace maps are continuous for the supplied trace*

topologies. They induce mutually inverse homeomorphisms of the fixed-endpoint scoped trace quotients. For binary observable trace topologies, continuity of either substitution follows if all substituted primitives have the same length k and their binary realizations equal the corresponding source primitive paths exactly.

Proof. Induction proves preservation of scoped rewrites and extends the primitive roundtrip witnesses to all traces. Quotient descent gives inverse maps and the quotient criterion gives their continuity. For the binary specialization, induction gives

$$\ell(Mp) = k\ell(p), \quad |Mp|_{\text{bin}} = |p|_{\text{bin}}.$$

The output observation is thus the continuous function $(n, \delta) \mapsto (kn, \delta)$ of the input observation. Apply this argument in both directions, with their respective expansion lengths. $\square$

The hypotheses concern the actual substitutions; no quotient homeomorphism is an input. The general criterion is quotient-topology infrastructure. The binary specialization derives continuity from primitive data, but its uniform-length and exact-timing conditions are restrictive. For the flat full-word topology, the following criterion instead permits variable-length substitutions, including empty words.

Theorem 5.3 (presentation invariance from primitive word codes). *Let M, N be endpoint-preserving finite-trace substitutions between presentations over the same space X . Suppose they preserve named rules and their primitive roundtrips are scoped rewrites to the original primitives. Suppose also that the two maps sending a primitive to the full signed word of its substituted trace are continuous into the coproduct of finite word spaces. Then they induce homeomorphisms of the flat trace-sensitive scoped quotients at fixed endpoints, of global arrow quotients, and of the final composable-pair domains. The pair maps commute with the two arrow projections. For discrete primitive alphabets the continuity hypotheses hold automatically.*

Proof. On each finite input stratum, substitution concatenates finitely many output words. Continuity of the primitive word codes makes this operation continuous; the discrete length coordinate isolates each output-length stratum. Equal-slot execution is continuous on the compatible finite-product strata, including the empty stratum X . Thus the output full word, endpoints, and realized path are continuous functions of the input source and full word. Choosing that realized path as the coherent representative gives a continuous map of the global carriers. Restricting its product to matching endpoints gives the continuous map of actual composable representative subspaces. Induction extends the primitive roundtrip rewrites to arbitrary traces. Quotient descent therefore gives inverse continuous maps in all three cases. The pair projection equations already hold on representatives. $\square$

The continuity conclusion is derived from primitive data; continuity of the whole substitution or a quotient homeomorphism is not assumed. These homeomorphisms preserve endpoints and scoped computation. Compatibility with the geometric homotopy comparison requires the additional condition that each substituted primitive realization is homotopic to the original path; `WordSubstitution.map_flat_homotopic` verifies its extension to all traces. Neither statement identifies final and ordinary composable topologies.

Proposition 5.4 (a redundant generator erases an observable distinction). *There is an algebraically conservative abbreviation move whose canonical fixed-endpoint*

quotient equivalence has a discontinuous inverse for the equal-slot observable trace topologies.

Proof. Choose any based path α . The old discrete-label presentation has primitives e, f , realizing $\alpha, \alpha \star \alpha$, and no named rules. Length parity $\ell(p) \bmod 2$ is invariant under structural scoped rewrites: cancellation removes an even number of letters. It is continuous because length is an observed discrete coordinate, and therefore descends to a continuous quotient invariant with values 1 on $[f]$ and 0 on $[e; e]$.

Add a primitive a realizing $\alpha \star \alpha$ and the sole defining rule $a \simeq_{\mathcal{P}} e; e$. Inclusion and substitution $a \mapsto e; e$ preserve all named rules and have primitive roundtrips equal up to scoped rewriting. They consequently induce inverse algebraic quotient maps. In the expanded presentation, the one-letter traces a, f have identical flat observable coordinates. Their quotient classes are inseparable. They are distinct: signed count of f is preserved by all structural rules and by the defining rule, and takes values 0, 1. Elimination sends these inseparable classes to $[e; e], [f]$, whose old continuous parity values differ. If elimination were continuous, its composite with parity would send inseparable points to distinct points of a discrete space, a contradiction. $\square$

This checked statement concerns the canonical presentation equivalence; it does not claim nonhomeomorphism under every unrelated bijection. It needs neither a nontrivial geometric loop nor geometric completeness. Matching coherent-representative elimination transports this fixed-endpoint result to the corresponding coherent quotients. Algebraic presentation changes already have an extensive theory; Henry and Mimram, for example, study Tietze equivalence for monoid presentations [16]. The issue here is preservation of a topology that observes computational length and exact parametrized paths. Brazas instead uses free topological groups to obtain a topological-group topology on the fundamental group [1]. Our final composable domain retains the declared arrow quotient topology. No priority claim is made for the abbreviation example or substitution criterion.

Hardy and Morris characterize free topological groupoids by extension of continuous graph morphisms, with composition continuous on the ordinary composable subspace [14]. Our construction adds declared rewrites and exact geometric observations and initially gives composable classes a final topology. Identifying it with their universal construction would require a separate argument. Michael's biquotient maps include open surjections and are stable under products and restriction over codomain subspaces [21]. They suggest a broader sufficient condition for ordinary/final agreement than the open-arrow criterion. That generalization is not proved here; product closure should not be confused with stronger characterizations that require additional hypotheses on the range.

6. GEOMETRIC COMPARISON AND COMPLETE PRESENTATIONS

Let H_{geo} be the quotient of coherent traces by equality of endpoints and endpoint-fixed homotopy of chosen geometric representatives, and write

$$h : T \longrightarrow H_{\text{geo}}$$

for the quotient map. Define the final composable geometric domain by

$$H_{\text{geo}}^{(2), \text{fin}} = T^{(2)} / \sim_{\text{geo}}^{(2)}, \quad (u, v) \sim_{\text{geo}}^{(2)} (u', v') \iff h(u) = h(u') \text{ and } h(v) = h(v').$$

The weighted concatenation invariance lemma makes representative-level concatenation descend to a well-defined composition on $H_{\text{geo}}^{(2),\text{fin}}$, and the quotient topology makes it continuous. The zero-weight hypotheses of the lemma hold because a coherent representative of an empty trace is null-homotopic. Composing the coherent identity map with h gives a continuous identity map into H_{geo} , while coherent reversal preserves geometric equivalence and therefore descends continuously. Together, these operations equip H_{geo} with its final-domain groupoid structure. The realization comparison and its composable-domain map are

$$R_{\mathcal{P}} : G_{\mathcal{P}} \longrightarrow H_{\text{geo}}$$

and

$$R_{\mathcal{P}}^{(2)} : G_{\mathcal{P}}^{(2),\text{fin}} \longrightarrow H_{\text{geo}}^{(2),\text{fin}}, \quad ([u], [v]) \longmapsto ([u]_{\text{geo}}, [v]_{\text{geo}}).$$

The first map sends a scoped class to its geometric homotopy class. The second applies that map to each member of a composable pair.

Here H_{geo} contains only *represented* geometric classes: it is a quotient of T , and not of the whole ambient path space X^I . Surjectivity of $R_{\mathcal{P}}$ is therefore built into its target. It does not say that the presentation represents every ambient homotopy class. The induced map $H_{\text{geo}} \rightarrow \Pi_1^q(X)$ is a continuous injection, since representative projection is continuous and h is quotient. We do not assert that this injection is a topological embedding. The universal presentation supplies all ambient paths through its one-letter section; the finite-generator circle and torus examples instead establish coverage of all based-loop homotopy classes at their chosen base points.

Soundness makes $R_{\mathcal{P}}$ well-defined. Geometric completeness makes it injective: ambient homotopy introduces no extra equalities among represented traces. Because both quotients here come from the same carrier T , equality of their equivalence relations then makes the comparison a homeomorphism.

Theorem 6.1 (comparison morphism). *The map $R_{\mathcal{P}}$ is continuous, surjective on the geometric quotient, and preserves source, target, identity, and reversal. The induced pair map $R_{\mathcal{P}}^{(2)}$ is continuous, and the two maps commute with composition on the final domains. In particular, $R_{\mathcal{P}}$ is a continuous groupoid morphism for the final-domain structures.*

Proof. Realization soundness makes the map well-defined. By construction $h = R_{\mathcal{P}} \circ q$, and h is continuous for the quotient topology on H_{geo} . Since $q : T \rightarrow G_{\mathcal{P}}$ is quotient, the quotient criterion makes $R_{\mathcal{P}}$ continuous. Every geometric class has a coherent raw representative by definition, proving surjectivity. The operation equations follow from the recursive realization of reflexivity, concatenation, and reversal. The map $R_{\mathcal{P}}^{(2)}$ sends a final-domain composable pair to the corresponding final-domain geometric pair, and the weighted concatenation invariance lemma gives preservation of the induced compositions. The raw quotient map $T^{(2)} \rightarrow H_{\text{geo}}^{(2),\text{fin}}$ is continuous and, by realization soundness, constant on componentwise scoped equivalence. It therefore factors through $q_{\text{fin}}^{(2)}$, and the quotient criterion makes $R_{\mathcal{P}}^{(2)}$ continuous. $\square$

Definition 6.2 (geometric completeness). The presentation is geometrically complete if, for all coherent representatives u, v , equality of their geometric homotopy

classes implies scoped equivalence:

$$R_{\mathcal{P}}(q(u)) = R_{\mathcal{P}}(q(v)) \implies u \approx v.$$

Theorem 6.3 (faithfulness and homeomorphism). *The comparison $R_{\mathcal{P}}$ is faithful, equivalently injective on arrows because its object map is the identity, if and only if $\mathcal{P}$ is geometrically complete. If $\mathcal{P}$ is geometrically complete, then*

$$R_{\mathcal{P}} : G_{\mathcal{P}} \xrightarrow{\cong} H_{\text{geo}}$$

is a homeomorphism on arrow spaces and an isomorphism of abstract groupoids.

Proof. If the comparison is injective, equality of the geometric homotopy classes of two coherent representatives gives equality of their scoped classes. This is precisely geometric completeness. Conversely, completeness turns equality of comparison images into equality of scoped classes. The comparison is always surjective and continuous. Under completeness, let $h : T \rightarrow H_{\text{geo}}$ be the geometric quotient map. If $h(u) = h(v)$, completeness gives $u \approx v$, hence $q(u) = q(v)$. Therefore q factors uniquely through a map

$$S : H_{\text{geo}} \longrightarrow G_{\mathcal{P}}, \quad S \circ h = q.$$

Since h is quotient and q is continuous, S is continuous. The two composites are identity maps by quotient induction. $\square$

In the trace-sensitive version, $G_{\mathcal{P}}$ and H_{geo} both carry the quotient topologies from T_{tr} . The homeomorphism always compares two quotients of the same carrier.

Remark 6.4 (Two distinct obligations). Geometric completeness concerns *which arrows are equal*: it compares the scoped and geometric equivalence relations. Product-quotient compatibility concerns *which composable-pair sets are open*: it compares the final and ordinary domain topologies and permits continuity of multiplication to pass to the ordinary pullback. The former does not imply the latter. The universal presentation is geometrically complete for every X , yet in the Hawaiian-earring example of Proposition 4.9 multiplication on its observable based-fiber quotient is discontinuous for the ordinary product topology.

Definition 6.5 (normal-form certificate). A normal-form certificate for $\mathcal{P}$ consists of a set NF of normal-form codes, a representative map $\nu : \text{NF} \rightarrow T$, and a normalizer $N : T \rightarrow \text{NF}$ such that, for all coherent representatives u, v ,

- (a) $u \approx \nu(N(u))$; and
- (b) $h(u) = h(v)$ implies $N(u) = N(v)$.

Clause (a) requires a scoped derivation to the chosen normal representative. Clause (b) requires geometrically equivalent representatives to receive the same code; equivalently, representatives with different codes lie in different geometric classes. A code records only the normal form; it stores neither a chosen homotopy nor a proof of normalization. No further condition on the representatives is needed, because equal codes have equal images under ν .

Theorem 6.6 (normal-form completeness criterion). *A normal-form certificate implies geometric completeness. Therefore the realization comparison is faithful and is a homeomorphism on arrow spaces.*

Proof. Assume $h(u) = h(v)$. Clause (b) gives $N(u) = N(v)$. Combining this with clause (a) for u and the symmetric form of clause (a) for v yields

$$u \approx \nu(N(u)) = \nu(N(v)) \approx v.$$

This is geometric completeness, and the faithfulness and homeomorphism claims follow from the comparison theorem. Figure 6 shows the two obligations separately. Neither N nor ν needs to be continuous. The certificate is a set-level object, and the topological conclusion comes from quotient factorization. $\square$

$$\begin{array}{ccc}
 h(u) = h(v) & & \\
 \downarrow \text{(b) semantic separation} & & \\
 N(u) = N(v) & & \\
 \downarrow \text{apply } \nu & & \\
 u \xrightarrow[\text{(a)}]{\approx} \nu(N(u)) = \nu(N(v)) \xrightarrow[\text{(a), symmetry}]{\approx} v & &
 \end{array}$$

FIGURE 6. A normal-form certificate proves completeness by two different means: syntactic normalization (a) connects each trace to its chosen normal representative; semantic separation (b) forces the two codes to agree. The bottom row is then scoped transitivity, yielding $u \approx v$. The arrows are logical steps and assume nothing about continuity of N or ν .

Corollary 6.7 (fiberwise normal-form completeness). *For fixed endpoints $a, b \in X$, the same definitions and proof apply to $T(a, b) \subseteq T$. In particular, a normal-form certificate on $T(a, a)$ proves geometric completeness of the based-loop comparison on that fiber.*

Proof. All endpoint maps and the scoped relation preserve fixed endpoint fibers, so the two normal-form clauses and the proof of the criterion restrict to $T(a, b)$ without change. $\square$

7. THE UNIVERSAL PRESENTATION

For every topological space X , take the primitive step space to be the continuous interval path space X^I . Its endpoint maps are evaluation at 0 and 1. Declare two parallel traces to be named generators exactly when their realized paths are endpoint-fixed homotopic. Thus every continuous path γ is itself available as a one-letter primitive trace. If two realizations are endpoint-fixed homotopic, that homotopy is literally one of the declared universal rules. The universal presentation is therefore complete in one step: the ambient homotopy relation is built into the presentation instead of being derived from a normalizer.

The two lemmas below treat both representative topologies explicitly. Without a superscript, $T_{\mathcal{U}_X}$ and $G_{\mathcal{U}_X}$ refer to the observable choice.

Lemma 7.1 (coherent-path quotient identification). *For the universal presentation, the projection*

$$\pi : T_{\mathcal{U}_X} \longrightarrow X^I, \quad \pi(p, \gamma) = \gamma,$$

is a continuous quotient map. Consequently, the quotient of $T_{\mathcal{U}_X}$ by endpoint-fixed homotopy of the chosen representative is canonically homeomorphic to the usual quotient-topologized path-class space

$$\Pi_1^q(X) = X^I / \simeq_{\partial I},$$

with the endpoint maps retained.

Proof. For T_{obs} , continuity of π is immediate from the fifth coordinate of κ_{obs} . For T_{tr} , it is the restriction of the second projection $\text{Tr}_{\mathcal{P}} \times X^I \rightarrow X^I$. Every $\gamma \in X^I$ has the coherent representative

$$\sigma(\gamma) = (\langle \gamma \rangle, \gamma),$$

where $\langle \gamma \rangle$ is the one-letter universal trace and the coherence is the constant endpoint-fixed homotopy. The map σ is continuous on the one-letter coproduct component for T_{tr} ; it is also continuous for T_{obs} because its observable code is built continuously from γ . Since $\pi\sigma = \text{id}$, if $\pi^{-1}(U)$ is open then $U = \sigma^{-1}(\pi^{-1}(U))$ is open. Thus π is quotient. The general quotient-factorization lemma applied to the homotopy relation now induces mutually inverse continuous maps between the two displayed quotient spaces. $\square$

Lemma 7.2 (fiberwise coherent-path quotient). *Fix $a, b \in X$, and let*

$$X^I(a, b) = \{\gamma \in X^I : \gamma(0) = a, \gamma(1) = b\}$$

carry its subspace topology. For $$ = tr, obs, define the coherent carrier with these fixed endpoints*

$$T_{\mathcal{U}_X}^*(a, b) = \{(p, \gamma) \in T_{\mathcal{U}_X}^* : s(p) = a, t(p) = b\}$$

with its own subspace topology, and define $G_{\mathcal{U}_X}^(a, b)$ to be its quotient by scoped equivalence, with the quotient topology. The restricted projection*

$$\pi_{a,b}^* : T_{\mathcal{U}_X}^*(a, b) \longrightarrow X^I(a, b)$$

is quotient. Hence $G_{\mathcal{U}_X}^(a, b)$ is canonically homeomorphic to the endpoint-fixed homotopy quotient of $X^I(a, b)$.*

Proof. The projection is continuous for either representative topology. Its one-letter section

$$\sigma_{a,b}(\gamma) = (\langle \gamma \rangle, \gamma)$$

lands in the fixed endpoint fiber and is continuous by the same coordinate argument as in Lemma 7.1. If $(\pi_{a,b}^*)^{-1}(U)$ is open, then $U = \sigma_{a,b}^{-1}((\pi_{a,b}^*)^{-1}(U))$ is open. Thus the restricted projection is quotient without appealing to a restriction theorem for the global projection. In the universal presentation, scoped equivalence on this fiber is exactly endpoint-fixed homotopy, so quotient factorization gives the claimed homeomorphism. $\square$

Two ingredients are needed for the universal identification (Figure 7). The continuous one-letter section makes representative projection a quotient map. The universal rewrite rules then identify scoped equivalence with endpoint-fixed homotopy. A section alone would not identify an arbitrary scoped quotient with the ambient path-class space; the two relations must agree as well.

Theorem 7.3 (universal completion). *For the universal presentation $\mathcal{U}_X$, scoped rewrite equality of parallel traces is equivalent to endpoint-fixed homotopy of their realizations. Consequently,*

$$G_{\mathcal{U}_X} \cong \Pi_1^q(X)$$

as abstract groupoids, and the comparison is a homeomorphism on arrow spaces.

Proof. Soundness is the general soundness theorem. Conversely, an endpoint-fixed homotopy is itself one of the named universal generators, so it is a scoped rewrite in one step. The equivalence of quotient relations follows after including endpoint equality. The coherent-path quotient identification lemma supplies the missing topological identification with $\Pi_1^q(X)$, and the geometric completeness theorem then gives the homeomorphism and groupoid isomorphism. $\square$

Corollary 7.4 (collapse of the two universal topologies). *For the universal presentation, the comparison map from Proposition 2.4*

$$J_{\mathcal{U}_X} : G_{\mathcal{U}_X}^{\text{tr}} \longrightarrow G_{\mathcal{U}_X}^{\text{obs}}$$

is a homeomorphism. More precisely, both spaces are canonically homeomorphic to $\Pi_1^q(X)$, and these homeomorphisms identify $J_{\mathcal{U}_X}$ with the identity on path classes.

Proof. The coherent-path quotient identification lemma applies to both representative topologies and identifies each universal scoped quotient with $\Pi_1^q(X)$. The quotient maps are induced by the same representative realization γ , so the two identifications commute with the identity on path classes. Hence the comparison map in Proposition 2.4 is the composite of one homeomorphism with the inverse of the other, and is therefore a homeomorphism. $\square$

$$\begin{array}{ccc} X^I & \xrightleftharpoons[\pi]{\sigma} & T_{\mathcal{U}_X}^* \\ q_I \downarrow & & \downarrow q_* \\ \Pi_1^q(X) & \xrightleftharpoons[\bar{\pi}]{\bar{\sigma}} & G_{\mathcal{U}_X}^* \end{array}$$

$$\pi\sigma = \text{id}_{X^I}, \quad G_{\mathcal{U}_X}^{\text{tr}} \cong \Pi_1^q(X) \cong G_{\mathcal{U}_X}^{\text{obs}}.$$

FIGURE 7. The universal one-letter section supplies the topology, and the universal rules supply the equivalence relation. Here $*$ $\in$ $\{\text{tr}, \text{obs}\}$ and $\sigma(\gamma) = (\langle \gamma \rangle, \gamma)$; q_I is the homotopy quotient map. The lower maps are induced by σ and π . The upper maps are continuous, with $\pi\sigma = \text{id}$, so π is quotient. Agreement of scoped and homotopy relations makes the lower maps mutually inverse homeomorphisms. Under these identifications, $J_{\mathcal{U}_X}$ is the identity on path classes.

Proposition 7.5 (the universal path projection). *Let $q_I^{\mathcal{U}} : X^I \rightarrow G_{\mathcal{U}_X}^{\text{obs}}$ send a path to the class of its one-letter coherent representative. This map is quotient. If $q_I^{\mathcal{U}}$ is open, the universal presentation has product-quotient compatibility, and multiplication is continuous on the ordinary pullback of its arrow space.*

Proof. The geometric-path projection $r : T_{\mathcal{U}_X}^{\text{obs}} \rightarrow X^I$ is continuous and has the continuous one-letter section σ . Every coherent representative has the same scoped class as the one-letter representative of its geometric path. Thus $q_T = q_I^{\mathcal{U}} \circ r$ and $q_I^{\mathcal{U}} = q_T \circ \sigma$. The first identity and the quotient property of q_T make $q_I^{\mathcal{U}}$ quotient.

On composable pairs, $r \times r$ again has a continuous section, so its restriction to the endpoint pullbacks is quotient. If $q_I^{\mathcal{U}}$ is open, its square remains an open quotient after restriction to the full inverse image of composable arrow pairs. The composite is the ordinary pair map for $\mathcal{U}_X$. The ordinary/final comparison theorem now gives product-quotient compatibility and continuity of multiplication. $\square$

Theorem 7.6 (a positive geometric class). *Let X be locally path-connected and semilocally simply connected. The universal observable presentation $\mathcal{U}_X$ has product-quotient compatibility. Its scoped quotient is therefore a topological groupoid for the ordinary pullback topology. For every $x \in X$, the based arrow fiber $G_{\mathcal{U}_X}(x, x)$ with the subspace topology from the global arrow space is discrete, and its multiplication is continuous on the ordinary product.*

Proof. Holkar, Hossain, and Kulkarni prove that the path-class projection $q_I : X^I \rightarrow \Pi_1^q(X)$ is open under these hypotheses [18, Corollary 3.7]; their Theorem 3.9 also identifies the resulting ordinary topological groupoid. The companion Lean proof makes the openness argument explicit. Let W be open in X^I , and choose $\gamma \in W$. For sufficiently small $\varepsilon > 0$, the path

$$t \mapsto \gamma(\max(0, \min(1, (1 + 2\varepsilon)t - \varepsilon)))$$

still belongs to W . It is constant near both endpoints. Choose a finite compact-open subbasic neighborhood of this reparametrized path inside W . Intersecting the target sets that contain $\gamma(0)$ gives an open set U around $\gamma(0)$; do the same at $\gamma(1)$ to obtain V . Any paths $\alpha : a' \rightarrow \gamma(0)$ and $\beta : \gamma(1) \rightarrow b'$ whose ranges lie in U and V can be inserted into the two constant tails. The resulting path remains in W and is homotopic relative to its endpoints to $\alpha\gamma\beta$. The latter claim follows by a piecewise linear change of speed.

Now suppose δ represents the same path class as γ . The finite null-homotopy ladder, using local path-connectedness and semilocal simple connectivity, gives a compact-open neighborhood N of δ . Each $\zeta \in N$ is homotopic to $\alpha\delta\beta$ for endpoint connectors with ranges in U and V . Replacing δ by γ preserves that homotopy class, and the absorption construction gives a representative in W . Thus N lies in the saturation of W . The saturation is open, so the quotient map q_I is open.

The arrow-space homeomorphism of Theorem 7.3 identifies q_I with $q_I^{\mathcal{U}}$. Proposition 7.5 therefore gives product-quotient compatibility and continuous ordinary multiplication.

The based arrow fiber is carried by that homeomorphism to the subspace of $\Pi_1^q(X)$ with both endpoints x . This subspace has the usual based-loop quotient topology: the based-loop space is the full inverse image of the fiber under the open map q_I , so the restricted map is still open and quotient. The one-letter section gives the same quotient topology from the coherent based fiber. Local path-connectedness and semilocal simple connectivity make the quotient fundamental group discrete [18, Corollary 3.10(iii)]. Its product is therefore discrete, and multiplication is continuous. $\square$

Corollary 7.7 (totally disconnected spaces). *If X is totally disconnected, $q_I^{\mathcal{U}}$ is a homeomorphism. The universal presentation has product-quotient compatibility and continuous multiplication on the ordinary composable-pair space. This conclusion does not require local path-connectedness.*

Proof. The interval is connected, so every continuous path in X is constant. Two such paths with the same scoped class have the same source and are therefore equal. Thus $q_I^{\mathcal{U}}$ is injective. It is quotient by Proposition 7.5, hence a homeomorphism. The remaining claims follow from the resulting openness of $q_I^{\mathcal{U}}$ and the open-projection criterion. $\square$

The Hawaiian earring in Proposition 4.9 is locally path-connected and fails the semilocal hypothesis at its common base point. These examples show how the ordinary-pullback obstruction behaves in geometric presentations. The positive groupoid theorem above is an application of the published compact-open quotient result; the presentation-level comparison is the contribution here.

8. CIRCLE AND TORUS PRESENTATIONS

These examples use traces whose endpoints are the chosen base point. Throughout this section, $G_{\mathcal{P}}(a, a)$ means the quotient of the coherent based-loop carrier $T(a, a)$ with its own quotient topology. It is not assumed to have the subspace topology inherited from the global arrow quotient. We first reduce words by the scoped rules, then use winding numbers to distinguish the resulting normal forms.

8.1. The additive circle: a finite generator and its completion. Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ carry the quotient topology. The minimal presentation has one oriented primitive step a at the base point 0, with realization

$$a(t) = [t] \in \mathbb{T}.$$

The formal reversal supplies a^{-1} ; no integer-indexed primitive step is assumed. A based trace is therefore a finite signed word in a and a^{-1} . No additional cancellation generator is needed: the structural inverse coherences in Definition 2.12 instantiate to $a; a^{-1} \simeq_{\mathcal{P}} 1_0$ and $a^{-1}; a \simeq_{\mathcal{P}} 1_0$, and are closed under the scoped congruence.

For a signed word w , let $\exp(w)$ be the number of positive letters minus the number of negative letters. Let c_n be the canonical word consisting of n copies of a when $n \geq 0$ and $-n$ copies of a^{-1} when $n < 0$.

Lemma 8.1 (finite-generator reduction). *Every signed word w has a finite scoped derivation*

$$w \simeq_{\mathcal{P}} c_{\exp(w)}.$$

If two canonical words have endpoint-fixed homotopic realizations, then their exponents are equal. In particular, scoped-equivalent canonical words have equal exponents by soundness.

Proof. Scan the word from left to right while retaining a reduced signed word. If the next sign agrees with the last retained sign, append it. If the signs differ, delete the adjacent inverse pair using the corresponding structural coherence inside the surrounding prefix and suffix, using concatenation congruence. Each deletion strictly decreases the word length, so the procedure terminates. A terminal reduced word has no adjacent sign change and hence consists entirely of positive letters or entirely of negative letters. The integer $\exp$ is unchanged by every deletion, so the

terminal word is exactly $c_{\exp(w)}$. The same invariant is the endpoint of the lift to $\mathbb{R}$; therefore endpoint-fixed homotopy of canonical words forces equality of their exponents. $\square$

For example, the cancellation mechanism gives the concrete scoped reduction

$$a; a^{-1}; a^{-1}; a; a \simeq_{\mathcal{P}} a^{-1}; a; a \simeq_{\mathcal{P}} a.$$

The first step cancels the initial $a; a^{-1}$ pair and the second cancels the remaining $a^{-1}; a$ pair.

The reduction lemma is the normal-form certificate from Section 6: the reduction derivation gives clause (a), and the lifted endpoint gives clause (b). Thus the minimal circle presentation is geometrically complete on its based loop fiber.

It is also useful to package the derived integer normal forms as a completed presentation with a primitive step λ_n for each $n \in \mathbb{Z}$,

$$\lambda_n(t) = [tn].$$

Its named rules are the derived integer rules

$$(n; m) \simeq_{\mathcal{P}} (n + m), \quad 1_0 \simeq_{\mathcal{P}} 0, \quad (n)^{-1} \simeq_{\mathcal{P}} (-n).$$

This integer-indexed presentation packages normal forms already derived from the single generator. The proof of the finite-generator classification does not use it.

Let L be the endpoint-fixed homotopy quotient of the based circle loop space, with its quotient topology. The map sending a loop class to its winding number and the map sending n to the standard loop class are inverse maps. Under the completeness conclusion below, write

$$S_0 : L \longrightarrow G_{\mathcal{P}}(0, 0)$$

for the inverse of the realization comparison on the based-loop fiber.

Theorem 8.2 (circle winding classification). *The finite-generator scoped presentation is geometrically complete on the based loop fiber, and winding number is a group isomorphism and a homeomorphism*

$$L \cong \mathbb{Z},$$

where $\mathbb{Z}$ is discrete. Moreover, the inverse realization comparison

$$S_0 : L \xrightarrow{\cong} G_{\mathcal{P}}(0, 0)$$

is a homeomorphism. In particular the scoped presentation is nondegenerate: the images of the zero and one winding loops are distinct.

Proof. A based loop has a unique lift to $\mathbb{R}$ starting at 0. Its endpoint is an integer, and endpoint-fixed homotopy preserves that integer. Straight-line interpolation of the lift with $t \mapsto tn$ shows that every loop of winding n is homotopic to λ_n . The lift of a concatenation is the first lift followed by a translate of the second, so winding is additive. Thus winding identifies L with $\mathbb{Z}$ as groups.

The preceding reduction lemma sends every trace to c_n , where n is its winding number, so the normal-form criterion gives geometric completeness. Explicitly, S_0 sends a loop class of winding n to $q(c_n, \lambda_n)$. Every scoped class has this form by normalization, and soundness prevents different windings from giving the same scoped class. Hence S_0 is inverse to realization. In particular, the zero and one winding classes are distinct.

It remains to prove continuity of S_0 . Give $\mathbb{T}$ its standard quotient metric. If two based loops γ, δ are uniformly less than $1/2$ apart, their difference has a unique continuous representative $d : I \rightarrow (-1/2, 1/2)$, with $d(0) = d(1) = 0$. Then $\gamma(t) + [u d(t)]$ joins them relative to endpoints. Compact-open and uniform topologies agree here since I is compact and $\mathbb{T}$ is metric. Winding is therefore locally constant and descends continuously from L to discrete $\mathbb{Z}$. Choosing (c_n, λ_n) for each n gives a continuous map from discrete $\mathbb{Z}$ to the coherent carrier and then to its scoped quotient. This constructs S_0 continuously; projection to the chosen loop supplies its continuous inverse on the fixed-endpoint quotient.

To check the topology inherited by $G_{\mathcal{P}}(0, 0)$ from the *global* scoped arrow space, consider a coherent raw representative with source x and chosen geometric loop γ . Every trace in this finite presentation has equal source and target. Define

$$W(x, \gamma) = \text{wind}(t \mapsto \gamma(t) - x).$$

Translation of a variable-basepoint loop to zero is continuous in the compact-open topology. The observable raw topology includes both x and γ , and winding is locally constant on based loops, so W is continuous on the full raw carrier. Scoped equivalence preserves the source and gives an endpoint-fixed homotopy of the chosen loops. Thus W descends continuously to the global arrow quotient. On the based subspace it is ordinary winding. Normal-form completeness makes this restriction injective, while the classes of (c_n, λ_n) make it surjective. Its inverse is continuous because $\mathbb{Z}$ is discrete. This proves the claimed homeomorphism for the actual subspace topology without restricting a quotient map. The based loop-class quotient is discrete; the loop space itself need not be. $\square$

8.2. The torus and its two generators. Let

$$\mathbb{T}^2 = (\mathbb{R}/\mathbb{Z}) \times (\mathbb{R}/\mathbb{Z})$$

with base point $(0, 0)$. Use two finite oriented generators

$$a(t) = ([t], [0]), \quad b(t) = ([0], [t]).$$

The only named rule is the commuting square

$$a; b \simeq_{\mathcal{P}} b; a;$$

inverse cancellations for a and b are structural, as for the circle. The commuting square is sound: it is the projection modulo $\mathbb{Z}^2$ of the evident square homotopy in $\mathbb{R}^2$. The other signed commutations are derived, rather than added as generators. For example, with unit insertions and structural cancellations displayed schematically,

$$\begin{aligned} a^{-1}; b &\simeq_{\mathcal{P}} a^{-1}; (b; a); a^{-1} \simeq_{\mathcal{P}} a^{-1}; (a; b); a^{-1} \simeq_{\mathcal{P}} b; a^{-1}, \\ a; b^{-1} &\simeq_{\mathcal{P}} b^{-1}; (b; a); b^{-1} \simeq_{\mathcal{P}} b^{-1}; (a; b); b^{-1} \simeq_{\mathcal{P}} b^{-1}; a, \\ b^{-1}; a^{-1} &\simeq_{\mathcal{P}} a^{-1}; b^{-1} \end{aligned}$$

and symmetry gives the reversed orientations. Thus closure under the scoped congruence supplies all four signed commuting rules.

For a signed word w , let $m(w)$ and $n(w)$ be its signed counts of a and b . Move every b -letter to the right using the commuting square and then perform the two cancellation algorithms from the circle. This is a finite terminating derivation

$$w \simeq_{\mathcal{P}} a^{m(w)}; b^{n(w)}.$$

The pair $(m(w), n(w))$ is invariant under every scoped rule. Each signed commutation strictly decreases the number of pairs in which a $b^{\pm 1}$ -letter occurs to the left

of an $a^{\pm 1}$ -letter. For a short worked instance, sorting and cancellation give

$$b; a; b^{-1}; a; b \simeq_{\mathcal{P}} a; b; b^{-1}; a; b \simeq_{\mathcal{P}} a; a; b = a^2; b.$$

The first step uses the commuting square and the second cancels $b; b^{-1}$; the winding pair is $(2, 1)$.

Lemma 8.3 (simultaneous and sequential torus representatives). *Let*

$$d_{m,n}(t) = ([mt], [nt])$$

be the simultaneous standard loop, and let $|a^m; b^n|$ be the equal-slot realization of the sequential word. Then

$$d_{m,n} \simeq_{\partial I} |a^m; b^n|.$$

Proof. Choose lifts of both loops to paths in $\mathbb{R}^2$ beginning at $(0, 0)$. The lift of $d_{m,n}$ is $t \mapsto (mt, nt)$, while the lift of the sequential equal-slot realization is piecewise linear. Both lifts end at (m, n) . The straight-line interpolation between these two lifted paths is therefore fixed at both endpoints. Composing it with the covering map $\mathbb{R}^2 \rightarrow \mathbb{T}^2$ gives the required endpoint-fixed homotopy. $\square$

Let $L_{\mathbb{T}^2}$ be the endpoint-fixed homotopy quotient of the based loop space of $\mathbb{T}^2$, with its quotient topology. Under geometric completeness, write

$$S_{\mathbb{T}^2} : L_{\mathbb{T}^2} \longrightarrow G_{\mathcal{P}}((0, 0), (0, 0))$$

for the inverse of the realization comparison on the based-loop fiber.

Theorem 8.4 (torus winding classification). *For the torus $\mathbb{T}^2$, the coordinate winding pair is a group isomorphism and a homeomorphism*

$$L_{\mathbb{T}^2} \cong \mathbb{Z} \times \mathbb{Z},$$

where $\mathbb{Z} \times \mathbb{Z}$ is discrete. The finite-generator scoped presentation is geometrically complete on the based loop fiber. Moreover, the inverse realization comparison

$$S_{\mathbb{T}^2} : L_{\mathbb{T}^2} \xrightarrow{\cong} G_{\mathcal{P}}((0, 0), (0, 0))$$

is a homeomorphism.

Proof. The universal cover is the product map

$$\mathbb{R}^2 \longrightarrow \mathbb{T}^2, \quad (x, y) \longmapsto ([x], [y]).$$

The lift of a based loop has endpoint $(m, n) \in \mathbb{Z}^2$, and endpoint-fixed homotopy preserves this pair. Conversely, the two coordinate projections of any based torus loop are circle loops; the circle lifting theorem gives endpoint-fixed homotopies to the standard loops of windings m and n . Taking their product gives a homotopy to the simultaneous loop $d_{m,n}$; Lemma 8.3 then gives a homotopy to the sequential realization $|a^m; b^n|$. Thus the coordinate winding pair classifies geometric loops, and it is additive because each coordinate winding is. The word reduction above gives the corresponding scoped derivation, so the normal-form criterion proves geometric completeness. Projection from coherent representatives to their chosen loops induces the continuous realization comparison. The local-constancy argument constructs its inverse continuously: each coordinate winding number is locally constant by the circle argument. The winding pair thus descends continuously to discrete $\mathbb{Z}^2$, and the map $(m, n) \mapsto (a^m; b^n, d_{m,n})$ is continuous because its domain is discrete. Lemma 8.3 ensures that each pair is coherent. Composing these maps with the scoped quotient map gives $S_{\mathbb{T}^2}$. For the topology inherited from the global scoped

arrow space, translate each variable-basepoint torus loop γ by its source x and take the two coordinate windings of $t \mapsto \gamma(t) - x$. This gives a continuous $\mathbb{Z}^2$ -valued map on the global quotient: translation and winding are continuous, and scoped soundness makes the value independent of the representative. Its restriction to the based subspace is bijective by the normal form above, with continuous inverse from discrete $\mathbb{Z}^2$. This proves the stated subspace homeomorphism. As for the circle, the loop space itself is not asserted to be discrete. $\square$

Figure 8 illustrates the lifted paths used in both proofs.

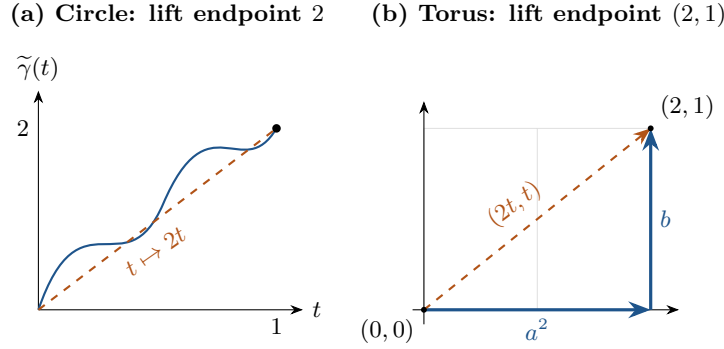


FIGURE 8. Winding separates normal forms because lifts remember endpoints. Left: a lift beginning at 0 and ending at 2 is straightened to $2t$. Right: the sequential word $a^2; b$ and the simultaneous loop have lifts with common endpoints $(0,0)$ and $(2,1)$; straight-line interpolation in $\mathbb{R}^2$ gives a homotopy after projection to the torus. Both plots are drawn in the covering spaces $\mathbb{R}$ and $\mathbb{R}^2$.

8.3. Products of complete based presentations. The torus sorting argument extends to a product without requiring normal forms for either factor. Let $\mathcal{P}$ be a sound scoped presentation on X with primitive steps E , and let $\mathcal{Q}$ be one on Y with primitive steps F . A product trace has horizontal letters (e, y') and vertical letters (x', f) , where $e \in E$, $f \in F$, $x' \in X$, and $y' \in Y$. Their realizations hold the other coordinate fixed. Lift each named rule of $\mathcal{P}$ at every y' , and each rule of $\mathcal{Q}$ at every x' . Add interchange around every primitive rectangle:

$$(e, y_0); (x_1, f) \simeq_{\mathcal{P} \boxtimes \mathcal{Q}} (x_0, f); (e, y_1), \quad e : x_0 \rightarrow x_1, \quad f : y_0 \rightarrow y_1.$$

The corresponding interchange for arbitrary factor traces $p : x_0 \rightarrow x_1$ and $q : y_0 \rightarrow y_1$ follows from these primitive squares:

$$H_{y_0}(p); V_{x_1}(q) \simeq_{\mathcal{P} \boxtimes \mathcal{Q}} V_{x_0}(q); H_{y_1}(p).$$

Here H_y and V_x lift entire traces while holding the other coordinate fixed. To derive the displayed relation, fix a vertical primitive step f and induct on p . The single-step case is the named rectangle rule; the unit case uses the groupoid unit laws. For a concatenation, paste the two previously derived squares along their common edge. For an inverse, reverse a square and cancel the resulting inverse pairs. Induction on q then gives the general trace relation by the same pasting and inverse steps. Write $\mathcal{P} \boxtimes \mathcal{Q}$ for the presentation with primitive rectangle rules. Its

rules are sound: the two paths are the two edge routes from (x_0, y_0) to (x_1, y_1) in the square $(u, v) \mapsto (\rho(e)(u), \rho(f)(v))$. The routes are homotopic relative to their endpoints inside the square.

Theorem 8.5 (product closure of based completeness). *Suppose $\mathcal{P}$ is geometrically complete on the based-loop fiber at $x \in X$ and $\mathcal{Q}$ is geometrically complete on the based-loop fiber at $y \in Y$. Then $\mathcal{P} \boxtimes \mathcal{Q}$ is geometrically complete on the based-loop fiber at (x, y) .*

Proof. Induct on a product trace. Reflection and single letters are already sorted after inserting a unit trace. For a concatenation, first sort both parts, then use the derived trace interchange to move the first part's vertical projection past the second part's horizontal projection. For an inverse, reverse the sorted trace and use interchange once more. Thus every trace rewrites to its horizontal projection followed by its vertical projection. Since the entire trace starts and ends at (x, y) , the horizontal projection is a loop at x , lifted at y , and the vertical word is a loop at y , lifted at x .

Take two coherent product loops whose geometric representatives are endpoint-fixed homotopic. Soundness and coherence let us replace each by its sorted realization. Projecting the resulting homotopy to X gives a homotopy between the two horizontal realizations: every vertical letter projects to a constant segment, which can be removed by an endpoint-fixed reparametrization homotopy. Projection to Y gives the corresponding homotopy between vertical realizations. Completeness of $\mathcal{P}$ and $\mathcal{Q}$ supplies scoped derivations between the respective factor words. Lift the first derivation at y and the second at x , then compose them with the two sorting derivations. This is a scoped derivation between the original product words, as required. $\square$

Iterating the theorem from the finite oriented circle presentation gives completeness of the corresponding product presentation of every n -torus on its based-loop fiber. The product construction includes a horizontal or vertical copy of each factor step at every point of the other factor, so its step space is generally infinite. It is distinct from the finite-generator torus presentation of Theorem 8.4. The Lean development checks sorting and based trace completeness for a presentation that names whole-trace interchange. It also derives every such rule from primitive squares, proves equivalence of the two scoped rewrite relations, and transfers completeness to the primitive-only presentation.

8.4. A geometric nonabelian example: the two-circle rose. Let R be the quotient of two copies of $\mathbb{R}/\mathbb{Z}$ identifying their zero points, with basepoint o . It is homeomorphic to the two coordinate circles inside $(\mathbb{R}/\mathbb{Z})^2$. Let a and b be their standard once-around based loops. Give the two primitive labels the discrete topology and declare no named rewrites. Thus cancellation and the structural groupoid rules are the entire scoped presentation.

Write $F(a, b)$ for the free group on the two labels. Reading a signed trace as a group word gives $\text{ev}(p) \in F(a, b)$. Its reduced word is an actual normal form: structural scoped rewriting derives p equivalent to the trace reconstructed from that word. This algebraic normalization works for any two based primitive loops. The following geometric bridge uses the particular space R .

Theorem 8.6 (represented-loop completeness for the rose). *For based traces p, q in this presentation,*

$$p \simeq_{\mathcal{P}} q \iff \text{ev}(p) = \text{ev}(q) \iff |p| \simeq_{\partial I} |q|.$$

Both the trace-sensitive and equal-slot observable fixed-endpoint based scoped quotients are discrete. In particular, $a; b$ and $b; a$ are geometrically nonhomotopic.

Proof. The normal-form argument gives the first equivalence, and soundness gives the forward geometric implication. For the converse, construct a space E with vertices $g \in F(a, b)$ and one closed interval from g to gs for each $s \in \{a, b\}$. Its topology is the final topology of these closed intervals, with precisely the specified endpoint identifications. Map each interval to the corresponding primitive circle loop and each vertex to o .

Left multiplication acts continuously on E . The map to R is a quotient map: the finite union of the two unit intervals maps continuously and surjectively from a compact space to the Hausdorff rose, and the projection forgetting the free-group coordinate is open and surjective. Descent through the interval-gluing quotient preserves this conclusion. The fibers are exactly the left-multiplication orbits. This follows from the injective open-interval parametrization of each punctured circle and the fact that the two circles meet only at o .

The action has disjoint local translates. Interior points have open neighborhoods contained in a single edge. At a vertex v , extend the word distance $\|v^{-1}g\|$ affinely along each edge. This is a continuous function d_v on E . The triangle inequality for reduced-word length gives

$$\|v^{-1}w\| \leq d_v(z) + d_w(z).$$

Consequently the open sets $d_v < 1/3$ and $d_w < 1/3$ are disjoint when $v \neq w$, and translation carries one such neighborhood to the corresponding translated vertex neighborhood. The orbit map is therefore a covering map.

A finite signed word has an explicit lift starting at the identity vertex, obtained by traversing the corresponding oriented edges. Its endpoint is the vertex given by its free-group evaluation. The equal-slot realization is homotopic to the recursively concatenated realization, so homotopic trace realizations have lifts with the same endpoint. Vertex injectivity now gives $\text{ev}(p) = \text{ev}(q)$ and hence scoped equality. The two reduced words ab and ba differ.

The full-word based quotient is discrete because the primitive alphabet is discrete. For the observable quotient, the four signed primitive paths are distinct: the two circle retractions and their winding numbers distinguish them. Equal-slot restriction recovers each signed letter. Each fixed-length observation stratum is finite inside a Hausdorff path space, hence discrete, and the discrete length coordinate separates the strata. Coherent-representative elimination identifies the coherent scoped quotients homeomorphically with these trace-only quotients, so they too are discrete. $\square$

This proof is a formalized instance of the classical graph-covering method [15, Section 1.A]. Represented-loop completeness identifies the subgroup generated by the two primitive loops. A classification of every loop in R additionally needs a proof that the covering realization is simply connected; it is a separate obligation.

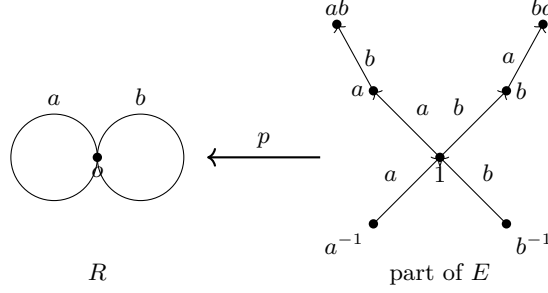


FIGURE 9. The geometric covering records word order in lifted endpoints. The two words ab and ba end at different vertices above the same basepoint. The pictured finite part belongs to the full interval-glued Cayley realization.

9. RELATION TO LOGIC AND REWRITING

In logical terms, a scoped derivation proves an equality in the presented groupoid $G_{\mathcal{P}}$, and soundness sends that equality to an equality of geometric homotopy classes. Normal forms can prove the converse for a particular presentation. In the universal presentation the converse holds by definition, since every endpoint-fixed homotopy is a named rewrite.

The semantic factorization theorem (Theorem 3.3) is the logical universal property of the construction. At this level, a scoped derivation is a finite proof object built from named rules and structural constructors; soundness is an induction principle on that derivation. Completeness is a separate theorem: whenever two represented paths are homotopic, their traces are related by the allowed rewrites. It is about equality of represented homotopy classes; individual homotopy witnesses are not represented.

The construction is compatible with the standard algebra of rewriting [5, 22, 19, 34, 13]: congruence, symmetry, and transitivity are already part of scoped equality. Additional named rules require homotopy witnesses, as in the definition of a presentation. Termination and confluence are not assumptions of the topological construction. They may help establish a normalizer, but the normal-form criterion still requires both a scoped normalization derivation and a proof that geometrically equivalent inputs have the same code. The finite-generator circle and torus reductions discharge those checks explicitly.

The quotient keeps trace classes and forgets the derivations between them, in the same way that coherence asks for a homotopy to exist and forgets which one. Computational data are used before quotienting; after quotienting, arrows are compared by ordinary equality. The construction does not produce a proof-relevant identity type or a higher groupoid of homotopy witnesses.

10. WHAT THE LEAN DEVELOPMENTS CHECK

Realization model boundary. The manuscript's trace carrier in Section 2 uses flat signed words with equal-duration slots. The earlier Lean modules use parenthesized traces and recursively apply Mathlib's half-and-half `Path.trans`. Their observable code retains the exact parametrized realization. Adding the flattened signed word

to that code does not remove the binary timing coordinate. Consequently, the existing Lean topology theorems concern the binary model. They must not be read as certificates of the equal-slot observable topology without a separate comparison theorem. Endpoint-fixed homotopy of the two realizations is sufficient for semantic soundness, but does not establish such a topological comparison. The binary development and its verified results are retained. A separate implementation now defines `GeometricTrace.flatRealize` on the same endpoint-indexed tree carrier. Each positive factor in a concatenation receives time proportional to its primitive length; zero-length factors are omitted. The theorem `flatRealize_slot` proves, including slot boundaries and reversal, that letter i occupies exactly $[i/n, (i+1)/n]$. The theorem `flatRealize_eq_of_flatWord` proves equality of parametrized realizations for trees with the same signed word. Thus parentheses are invisible to this realization. The theorem `flatRealize_homotopic_binary` supplies the endpoint-fixed homotopy bridge; it does not compare the two observable quotient topologies.

`EqualSlotTopology` defines separate fixed-endpoint and global observation topologies using this exact flat realization, without retaining a binary realization coordinate. The coherent carrier can be reused: `coherent_iff` proves that coherence with the binary and flat paths are equivalent propositions, and `scoped_sound` transfers rewrite soundness. The literal-word bridge now closes the two additional obligations. `LiteralWord.Word.normalization` reconstructs each tree from its composable signed word using only units, associativity, and reversal laws. An independently generated word relation uses adjacent-letter cancellation and the flattened named rules; `WordRwEq.ofTrace_iff` proves that it agrees with tree rewriting. Equality of geometric paths is never introduced as a rewrite rule.

The literal finite strata have $W_0 = X$ and the compatible positive signed tuples with their product-subspace topologies. Harmless Lean universe lifts are removed by explicit homeomorphisms. Canonical equal-slot reconstruction is continuous on every stratum. The coproduct topology is induced by $(s(p), p)$, where the second coordinate is the complete signed word: the source retains the basepoint of an empty word. The source coordinate is redundant on every positive stratum. The theorems `Word.sensitiveTopology_eq_tupleTopology` and `TotalWord.sensitiveTopology_eq_coproductTopology` then identify the full-word refinements with these genuine literal-word topologies. Neither result identifies a binary observable topology with a flat one.

Proposition 10.1 (binary timing can identify observable codes). *There is a three-generator based circle presentation with no named rewrite rules for which the observable quotients of the fixed-base carriers differ: the binary quotient is not T_0 , and the equal-slot flat quotient is discrete.*

Proof. Let $\alpha(t) = [t]$ in $\mathbb{R}/\mathbb{Z}$, and let the primitive labels a, b, c realize α , $\beta = \alpha \star \alpha$, and $\chi = \alpha \star \beta$, respectively, with $\star$ denoting ordinary half-and-half concatenation. The binary traces $p = (a; b); b$ and $q = c; (a; a)$ both realize $\chi \star \beta$. They have length three and can choose this same path as coherent representative, so their binary observable codes agree. Signed occurrence count of c is invariant under structural scoped rewrites and has values zero and one on these two traces. Their scoped classes are distinct. Every open set in the induced raw observable topology contains both chosen representatives or neither; the same holds for their classes by the definition of quotient topology. Hence that quotient is not T_0 .

For the flat construction, the six oriented primitive paths have distinct winding numbers $1, 2, 3, -1, -2, -3$. Restriction to each prescribed equal slot therefore recovers the signed letter. At each fixed length there are finitely many words, with distinct realizations in the Hausdorff compact-open path space. This finite image is discrete, so the fiber of each observed word is open. The discrete length coordinate separates the strata. A scoped-class preimage is a union of word fibers, even when the coherent path varies; thus the fixed-base flat quotient is discrete. This last assertion concerns the quotient of the based carrier, not an unproved subspace comparison with the global quotient. $\square$

The exact binary collision is implemented in `BinaryTimingCollision`, which proves failure of T_0 for both the fixed-base trace quotient and the global binary scoped quotient. Coherent-representative elimination transfers the fixed-base trace result to the matching coherent quotient. The actual circle winding computation, flat discreteness, and exclusion of every binary/flat observable homeomorphism are checked in `FlatObservableDiscreteness.BinaryCircle`.

Proposition 10.2 (fixed-endpoint coherent-representative elimination). *For a fixed realization model and endpoints, give traces a topology making realization continuous and coherent representatives the topology induced by (p, γ) . If scoped equivalence depends only on the trace, forgetting γ induces a homeomorphism between the coherent quotient and the trace-only scoped quotient.*

Proof. The projection $r(p, \gamma) = p$ is continuous, and $s(p) = (p, |p|)$ is a continuous section. Both maps preserve the quotient relations. We have $rs = \text{id}$, while $sr(p, \gamma)$ is scoped equivalent to (p, γ) because the trace is unchanged. Quotient descent therefore gives continuous mutually inverse maps. No geometric completeness hypothesis is needed. Apply the statement separately to each realization model and each chosen trace topology. $\square$

`CoherentRepresentativeElimination` implements this result for fixed endpoints with the binary observable trace topology and its full-word refinement. Its general declaration accepts any trace topology for which the same binary realization is continuous. `EqualSlotTopology` proves the matching result with the equal-slot canonical section, including the observable trace topology and its full-word refinement. These are separate fixed-endpoint homeomorphisms. `LiteralWord.TotalWord` extends matching equal-slot elimination and structural reconstruction to global arrows, and `LiteralWord.WordPair` extends them to the quotient of the actual endpoint-matched representative subspace. Both the observable and full-word versions are checked; the full-word version is also identified with the genuine literal-word coproduct. These results do not identify the ordinary pullback topology, a based subspace inside a global quotient, or a binary observable topology with a flat one.

The presentation-change milestone adds `ScopedWordInterpretation` and `ContinuousWordInterpretation`. The former checks primitive finite-trace substitutions, named-rule-to-scoped preservation, primitive roundtrip extension, and the algebraic `quotientEquiv`. The latter checks `WordSubstitution`, `mapTrace_length`, `mapTrace_realize`, and `continuous_binary`, and derives `ScopedWordInterpretation.quotientHomeomorph` for supplied trace topologies and `binaryQuotientHomeomorph` under the primitive conditions of Proposition 5.2. `RedundantGeneratorObservable.quotientEquiv` checks the conservative move

Result	Exact local declaration	Assumptions	Model
Binary elimination	<code>quotientHomeomorph</code>	Fixed endpoints; realization continuous for supplied trace topology; trace-only scoped relation	Binary
Timing collision	<code>realize_eq</code>	Any based path; three discrete labels with the stated realizations	Binary
Based failure of T_0	<code>fixed_not_t0</code>	Same three-label system; no named rewrite rules	Binary
Global failure of T_0	<code>not_t0</code>	Same three-label system; no named rewrite rules	Binary
Exact slot formula	<code>flatRealize_slot</code>	Endpoint-indexed trace; index $i < n$ and $t \in I$	Flat tree
Word independence	<code>flatRealize_eq_of_flatWord</code>	Fixed endpoints; equal signed words	Flat tree
Homotopy bridge	<code>flatRealize_homotopic_binary</code>	Any endpoint-indexed trace	Both
Flat elimination	<code>quotientHomeomorph</code>	Fixed endpoints; flat realization continuous for supplied trace topology	Flat tree
Flat discreteness	<code>BinaryCircle.flat_traceQuotient_discrete</code>	Circle paths with six distinct oriented realizations; fixed-base carrier	Flat

TABLE 1. Model-boundary verification map. The first row is in `CoherentRepresentativeElimination`; the next three are in `BinaryTimingCollision`, all under `ComputationalPaths.Path.GeometricTopology`. The next three are in `GeometricTrace`, and flat elimination is in `EqualSlotTopology`. The flat-discreteness row is in `FlatObservableDiscreteness`. The earlier eight formal rows passed Lean 4.32.0 kernel checks at source commit 404ac01, with Mathlib commit 81a5d257; only `propext`, `Classical.choice`, and `Quot.sound` occur in their axiom closures. Flat discreteness and the nonhomeomorphism conclusion are now formalized in `FlatObservableDiscreteness.BinaryCircle`, using winding numbers of the six actual oriented primitive paths. The earlier manifest `model-boundary-replay.json` at commit 131e4bf selected 19 actual declarations for direct NanoDa replay, accepted in the exact-head Quality run. The five earlier Comparator selections retain their own scopes. Later selections extend this coverage as described below.

of Proposition 5.4; `continuous_oldParity`, `new_classes_ne`, `new_inseparable`, and `not_continuous_inverse` check its topological failure for the fixed-endpoint flat-tree observable quotients. All these declarations passed the Lean 4.32.0 kernel at source commit d604b75, with the same pinned Mathlib and only the three standard axioms listed in Table 1. The expanded direct-replay manifest selects 37 actual declarations; its hosted acceptance must be checked on the exact published head. That earlier 37-declaration milestone included no general variable-length flat invariance, global presentation equivalence, or final-pair transport. The current

milestone adds those results under the primitive-word continuity hypotheses of Theorem 5.3.

The literal-word milestone adds the checked results in Table 2. All use Lean 4.32.0 with Mathlib 81a5d257; their axiom closures contain only `propext`, `Classical.choice`, and `Quot.sound`. The source snapshot is 0346c8e. The direct replay manifest selects the actual declarations; exact-head hosted acceptance is recorded with the published artifact, separately from local kernel checking.

Table 2: Literal-word verification map. Names are in `ComputationalPaths.Path.GeometricTopology.LiteralWord`. Every row has source commit 0346c8e.

Result	Exact declaration	Assumptions	Model
Structural normalization	<code>Word.normalization</code>	Any continuous scoped presentation; no named rule used	Both
Same-word rewriting	<code>Word.scopedEq_of_flatWord</code>	Parallel trees with equal signed words	Both
Independent relation	<code>WordRwEq.ofTrace_iff</code>	Adjacent-letter cancellation and flattened named rules	Literal/ tree
Stratum execution	<code>Strata.continuous_flatRealize_family</code>	Continuous primitive realization; genuinely composable tuples	Flat
Initial coproduct code	<code>Strata.totalWordTopology_eq_induced_code</code>	Empty basepoint retained; compatible finite tuple strata	Flat
Variable-length family	<code>Strata.continuous_flatRealize_of_flatWord</code>	Continuous endpoints and full signed word; typed compatibility	Flat
Fixed tuple comparison	<code>WordRwEq.tupleQuotientHomeomorph</code>	Matching full-word topology and scoped relation	Flat
Global comparison	<code>TotalWord.coproductQuotientHomeomorph</code>	Matching equal-slot coherent carrier and literal coproduct	Flat
Final-pair comparison	<code>WordPair.coproductQuotientHomeomorph</code>	Quotient of actual composable representative subspace	Flat
Discrete full-word quotient	<code>discrete_traceSensitiveQuotient</code>	Fixed endpoints and discrete labels; alphabet need not be finite	Flat

Table 3: Presentation invariance and separation verification map. Names are under `ComputationalPaths.Path.GeometricTopology`. Every row has source commit 0346c8e.

Result	Exact declaration	Assumptions	Model
Word substitution	<code>FlatWordSubstitution.continuous_substitution</code>	Continuous primitive word codes; arbitrary finite output lengths	Flat
Fixed invariance	<code>ScopedWordInterpretation.flatSensitiveQuotientHomeomorph</code>	Rule preservation; primitive roundtrip rewrites; continuous word codes	Flat

Result	Exact declaration	Assumptions	Model
Global invariance	<code>ScopedWordInterpretation.</code> <code>flatSensitiveGlobalHomeomorph</code>	Same hypotheses; common ambient space	Flat
Final-pair invariance	<code>ScopedWordInterpretation.</code> <code>flatSensitiveFinalPairHomeomorph</code>	Same hypotheses; actual matching-endpoint representative subspace	Flat
Projection compatibility	<code>ScopedWordInterpretation.</code> <code>finalPairMap_left</code> , <code>ScopedWordInterpretation.</code> <code>finalPairMap_right</code>	No ordinary product-quotient hypothesis	Flat
Abbreviation invariance	<code>RedundantGeneratorObservable.</code> <code>flatSensitiveGlobalHomeomorph</code> , <code>RedundantGeneratorObservable.</code> <code>flatSensitiveFinalPairHomeomorph</code>	Discrete old and new alphabets; the actual defining rule	Flat
Timing nonhomeomorphism	<code>FlatObservableDiscreteness.</code> <code>BinaryCircle.binary_flat_not_homeomorphic</code>	Actual three-label circle paths; fixed endpoints; no named rules	Both
Complete presentation	<code>CompleteDiscreteSeparation.</code> <code>complete</code>	All continuous paths as discrete labels; homotopy named rules	Both
Ordinary quotient recovery	<code>CompleteDiscreteSeparation.</code> <code>observableHomeomorph</code>	Any based topological space; compact-open path topology	Flat
Complete separation	<code>CompleteDiscreteSeparation.not_homeomorphic</code>	Ordinary based quotient explicitly assumed nondiscrete	Flat

The current direct-replay manifest selects 141 actual declarations (99 theorems and 42 definitions), extending the 99-declaration literal-word selection. Its source and dependency closure are audited against the manifest before NanoDa replay. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted. The exact published-head Quality run checks these declarations separately from the five retained Comparator selections. The rose development adds an actual geometric carrier and a checked Cayley-edge covering to the reduced-word calculation. It proves completeness between represented traces, geometric noncommutation, and discreteness of the two fixed-endpoint flat computational quotients. Classifying arbitrary rose loops remains a separate obligation: the covering realization still needs a checked simple-connectivity proof.

Table 4: Geometric rose verification map. Every name has prefix `ComputationalPaths.Path.GeometricTopology`. The source snapshot is `dbfe65e`, with Lean 4.32.0 and Mathlib 81a5d257. The listed results use only the three permitted standard axioms.

Result	Exact declaration	Assumptions	Model
Actual rose	<code>GeometricRose.</code> <code>axesHomeomorph</code>	Two additive circles, exactly their zero points glued	Geometric
Reduced normal form	<code>RoseReducedWords.</code> <code>normalization</code> , <code>RoseReducedWords.scoped_</code> <code>iff_encode_eq</code>	Any two based primitive loops; no named rewrites	Scoped algebra
Geometric covering	<code>CayleyRoseCovering.covering</code>	Actual interval-glued free-group edges; proved quotient and orbit fibers	Geometric

Result	Exact declaration	Assumptions	Model
Computed endpoint	<code>RoseLiftedWords.liftPath_trace_one</code>	Actual rose primitives and covering; identity starting vertex	Binary path
Trace completeness	<code>RoseLiftedWords.trace_complete</code> , <code>RoseLiftedWords.flat_trace_complete</code>	Based finite traces; actual endpoint-fixed homotopy	Binary and flat
Order separation	<code>RoseLiftedWords.ab_not_homotopic_ba</code>	Actual once-around primitive loops	Geometric
Signed observation	<code>RoseBasedTopology.signedPrimitive_injective</code>	The four actual oriented paths; winding pairs through retractions	Parametrized
Based quotients	<code>RoseBasedTopology.flatObservableHomeomorph</code> , <code>RoseBasedTopology.flatSensitiveHomeomorph</code>	Fixed endpoints; respective proved discrete quotient topologies	Flat
Tree prefix step	<code>CayleyReducedPrefix.outward_word_cases</code>	Actual reduced free-group words; no geometric assumptions	Algebraic

Two Lean developments accompany the paper. The parent artifact (Section 10.1) uses Lean 4.24.0 and Mathlib 4.24.0 and covers the constructions listed in Table 6. A smaller extraction, built with Lean 4.32.0, contains one bundled statement registered in Palomar (Section 10.2). The statements and proofs in this paper can be read without either implementation, and in both cases the checks cover the named declarations and nothing else.

10.1. The parent artifact.

Traces, soundness, and quotient operations. The implementation represents traces by the parenthesized inductive type `GeometricTrace`, where Section 2 uses flat signed words. The declarations `WeightedSlotReparam`, `balancedSlotReparam`, and `weightedConcatenationInvariant` relate the parametrizations up to endpoint-fixed homotopy, including the null-homotopy assumptions needed when a factor has zero weight. They do not identify observable topologies. The scoped-rewrite modules check soundness and the quotient operations. The comparison with Mathlib’s fundamental groupoid identifies the represented homotopy classes, checks preservation of identities, inverses, and composition, and proves that the induced map on final composable-pair domains is continuous.

Representative topologies. The module `TraceSensitiveTopologicalCompPath` flattens internal traces to signed words while retaining binary realization in its observable coordinate. It checks continuity of that binary realization, of the map to observable coordinates, and of the identity, reversal, and binary composition operations. The earlier quotient modules still use the observable topology by default.

The module `TraceSensitiveSeparation` checks the two-label separation mechanism on a finite code space: the one-letter trace codes differ, their observable codes agree, and the identity from the discrete trace topology to the indiscrete observable topology is continuous while its inverse is not. It also checks that the two circle labels realize the same nonconstant loop, proves the signed-count invariant for every structural scoped rewrite, and separates their scoped quotient classes. Their full observable codes agree. Signed count descends continuously to the trace-sensitive

quotient; the comparison to the observable quotient is continuous, but its inverse is not. Thus the topological separation in Example 2.8 is checked in Lean. The module `TraceSensitiveUniversalCollapse` checks the continuous one-letter section and the homeomorphism between the two universal quotient topologies. The abstract mechanism of Proposition 2.6 is checked in `CompleteDiscreteSeparation`. Its labels are all continuous interval paths, with a separate discrete label topology, and its named rules are exactly homotopies of parallel realized traces. The global geometric-completeness criterion is proved directly. For based loops at a fixed x , `observableHomeomorph` identifies the flat observable quotient with the ordinary quotient fundamental group, while `sensitive_discrete` proves discreteness of the full-word quotient for the possibly infinite alphabet. With nondiscreteness of the ordinary based quotient as an explicit hypothesis, `not_homeomorphic` excludes every homeomorphism between the two topologies. Matching coherent-representative versions are checked too. The harmonic archipelago and Fabel’s facts about its based quotient remain external mathematical inputs; no such facts are introduced as Lean axioms.

Circle and torus examples. The module `FiniteCircleTorusPresentation` uses one circle generator and two torus generators. It checks integer and integer-pair codes for traces, completion certificates, and soundness of the torus commuting square. The modules `BasedScopedTraceGroup`, `FiniteCircleScopedCompleteness`, and `FiniteTorusScopedCompleteness` prove normalization for the exact finite presentations. The circle proof uses only the structural groupoid laws; the torus proof adds its single named commuting square. Both prove based geometric completeness, instantiate the trace-topology comparison, and identify the based scoped quotients with discrete $\mathbb{Z}$ and $\mathbb{Z}^2$, respectively. Composing with the geometric winding homeomorphisms gives explicit homeomorphisms from the scoped based quotients to the ordinary circle and torus loop quotients. The separate torus development in `ComputationalPaths/Path/Topology/TopologicalTorusScoped.lean` includes the comparison between simultaneous and sequential loop representatives.

Hawaiian-earring transfer. The dedicated module checks that an externally supplied failure of the product-quotient property yields failure of ordinary-pair compatibility, and it transfers discontinuity of multiplication along the comparison homeomorphism. Fabel’s results enter as assumptions; the Lean development does not reprove them.

Build and archive. The root import file is `ComputationalPaths.lean`, and Table 6 in Appendix A lists the principal declarations. The core artifact, version 0.6.1+lean-only-v3, is archived on Zenodo under the version DOI [doi:10.5281/zenodo.21938980](https://doi.org/10.5281/zenodo.21938980); its concept DOI is [doi:10.5281/zenodo.21817207](https://doi.org/10.5281/zenodo.21817207) [31]. The source, including the finite-generator module, the torus simultaneous/sequential bridge, and the trace-sensitive topology module, is also in the GitHub repository. Build the project with `lake build`; the artifact README lists commands for building individual modules separately.

The checked modules contain no unfinished proofs and no custom axiom declarations. Declaration-level axiom inspection reports only Lean’s standard logical and quotient principles: propositional extensionality, classical choice, and quotient soundness. The development complements earlier work on computational paths and the Lean theorem prover [6, 25, 28].

10.2. The registered result. The extraction *TopologicalComputationalPaths* has a registered Palomar result, PALOMAR-2026-08-29-000005, version 1, published on 29 August 2026 [32]. It selects the single bundled declaration `TopologicalComputationalPaths.main_result` at the immutable commit

8254c40d0de03ff469c7c9c05087b8bc154e9a87,

built with Lean 4.32.0 and its checked-in dependency manifest. The selected statement is an `OrdinaryTopologyComparisonCertificate`, and Table 5 summarizes what it covers.

The certificate has three parts. The first gives the continuous bijection between final and ordinary pair spaces, the equivalent criteria for their topologies to agree, the compact-Hausdorff and discrete sufficient conditions, the consequence for ordinary composition, and the obstruction obtained from discontinuity. (The discrete field assumes that both $G_{\mathcal{P}}$ and its final domain are discrete; by Corollary 4.8 the first assumption implies the second.) The second part classifies based-loop homotopy classes of the circle and torus by integers and integer pairs. It checks the invariant, the standard representatives, the identity value, additivity under concatenation, and that the classification maps are mutually inverse. The homeomorphism statements of Theorems 8.2 and 8.4 are not part of it. The third part transfers the Hawaiian-earring obstruction under the hypotheses described below.

Part of the paper	Scope of the registered certificate
Final/ordinary comparison	Canonical continuous bijection; exact quotient/homeomorphism/topology criteria; sufficient conditions and ordinary-composition consequence.
Circle and torus	Additive classifications of the actual loop quotients, with explicit standard representatives and inverse laws.
Hawaiian earring	Concrete observable based fiber and quotient comparison; obstruction transfer conditional on <code>FabelHawaiianEarringFacts</code> .
Broader exposition	Not certified by this selection: the trace-sensitive refinement, the minimal finite-generator rewrite algorithms, the homeomorphism statements for the circle and torus, the explanatory homotopies and figures, and the manuscript as a whole.

TABLE 5. What the Palomar registration covers. Supporting code and mathematical exposition are listed apart from the selected conclusions.

The Hawaiian-earring field needs a caveat. In the registered statement, the topology `hawaiianObservableOpenFiberTopology` is induced by the geometric projection alone, and its one-letter section identifies its quotient with the standard loop quotient. This topology was defined for the registered based-fiber result; identifying it with the topologies of Section 2 would need a proof. The field `hawaiian_based_fiber` assumes `FabelHawaiianEarringFacts`, so the failure of the product-quotient property and the discontinuity of multiplication are hypotheses there, and the extraction does not reprove them. In the mathematical account, Proposition 4.9 takes them from Fabel’s published result [12].

The current working source after that immutable version equips the Hawaiian observable based carrier with the topology induced by its full observable code. The revised one-letter section and geometric projection are continuous, and the based quotient remains homeomorphic to the usual loop quotient. It also weakens the discrete sufficient condition in the selected certificate to discreteness of the arrow space alone, and checks continuity of both projections from the final composable domain. The present Lean source also checks the general open-arrow criterion and Proposition 7.5: the path-class projection is quotient, and its openness implies ordinary multiplication for the universal presentation. A direct Lean proof now supplies openness under the hypotheses of Theorem 7.6. The theorem itself remains credited to Holkar, Hossain, and Kulkarni.

On fixed endpoint quotients, Lean now identifies the universal based fiber with the ordinary based-loop quotient. It proves discreteness, a quotient pair map, and continuous ordinary multiplication under the semilocal hypothesis. It also identifies based paths with the endpoint-defined subspace of the compact-open path space and checks that an open global projection restricts to an open quotient onto the based-arrow subspace. For locally path-connected, semilocally simply connected spaces, the checked open-map theorem removes the remaining hypothesis from that comparison. The resulting global based-arrow subspace is homeomorphic to the ordinary based-loop quotient and is discrete. The module **UniversalTotallyDisconnected** proves the unconditional case of Corollary 7.7. Its projection is a homeomorphism, and ordinary composition is continuous. **EndpointVaryingLadder** establishes the local ladder step: every path has a compact-open neighborhood whose members differ by short endpoint connectors. **EndpointAbsorption** first gives the center path positive constant tails. It then inserts connectors confined to suitable endpoint neighborhoods and proves, by an explicit reparametrization, that the new path has the required concatenation homotopy class. The module **UniversalSemilocallySimplyConnected** uses both lemmas to show that the saturation of any compact-open set is open. It also records ordinary pair compatibility, continuous composition, and the global based-fiber homeomorphism as consequences. The integer-indexed scoped circle presentation has a checked normal form; its fixed endpoint quotient is homeomorphic to $\mathbb{Z}$, and a continuous winding-based trace choice identifies its trace-sensitive and observable quotient topologies. This integer-indexed certificate is distinct from the finite-generator circle presentation in Section 8. The current source also checks the general trace-section criterion, the universal trace collapse, a finite separation model, and the realized fundamental-groupoid arrow comparison. For products it defines the horizontal and vertical continuous step system, checks geometric soundness of primitive interchange, and proves sorting and based trace completeness for the presentation in Theorem 8.5. The module **ProductPrimitiveInterchange** derives the whole-trace rule from primitive rectangles, shows that the two presentations induce the same scoped rewrite relation, and transfers based completeness to the smaller presentation. For the finite-generator circle and torus, it checks the continuous winding-based choices of signed traces and proves the required based completeness for their exact scoped rules. The resulting based quotient homeomorphisms and trace-topology comparisons are unconditional. A comparison with the subspace fibers of their global arrow quotients is also checked: translation of a variable-basepoint loop to zero gives a continuous winding map on each global

scoped quotient, and finite based completeness makes its restriction to the actual based-arrow subspace injective. The resulting subspaces are homeomorphic to $\mathbb{Z}$ and $\mathbb{Z}^2$, respectively, and thus to the ordinary geometric loop quotients. The Lean source for these working results is available at `TopologicalComputationalPaths` commit `09b20abe0a4f69b361846c2466bff8528044fa93`. With the repository's Lean 4.32.0 toolchain and checked-in Mathlib manifest, `lake build` checks the development and all configured targets. The manuscript source, bibliography, and figures are in `paper/topological` and `paper/refs.bib`; the manuscript is not itself a theorem selected by Comparator. None of these working-source additions has been registered as a new Palomar version. The internal `comparator-roadmap.json` bundle selects a 24-field certificate containing the checked semilocal and totally disconnected universal cases, product results, finite global fibers, and trace-topology comparisons. Its Challenge imports the substantive development, and its Solution assembles the proved declarations. Comparator, NanoDa, and Lean's kernel replayed this bundle in CI. Palomar's current Challenge provenance rule excludes local imports, so the bundle is not a candidate for registration. A separate Mathlib-only Challenge in `comparator-registry-roadmap.json` selects the open-arrow ordinary-pair quotient, continuity transfer for multiplication, and the ordinary geometric circle and torus loop-quotient homeomorphisms. These four statements have not undergone Palomar's mechanical preflight or external review. The finite scoped global-fiber results remain outside that selection.

To reproduce the registered version 1 result, check out its pinned commit and run `lake build`. The selection file `comparator.json` pairs `Challenge.lean` with `Solution.lean`. The theorem placeholder in `Challenge.lean` states what is checked against the proof in `Solution.lean`, so it does not count as an unfinished proof. Later follow-up and preimage-solver development is outside this registered snapshot. Registration gives a versioned verification record for the selected formal statement and nothing more. It does not cover the prose or the illustrations, and it implies neither journal acceptance nor an endorsement of novelty.

11. CONCLUSION

Two questions come up when computational paths are interpreted geometrically, and this paper keeps them apart. The first is algebraic: which traces should count as equal? A scoped presentation answers with its declared rewrites and the groupoid laws. Soundness guarantees that equal traces realize homotopic paths, and the converse needs a completeness proof, which the normal-form criterion supplies for the based-loop circle and torus presentations. The product theorem extends based completeness from two presentations to their product, giving complete based-loop presentations for all finite tori. Their step spaces need not be finite; the finite-generator circle and two-torus presentations are separate examples. For the two-circle rose, the Cayley-edge covering supplies the geometric bridge: homotopic represented loops have equal lifted endpoints and hence equal reduced words. Its fixed-endpoint flat scoped quotients are discrete; the words ab and ba remain distinct even geometrically.

The second question is topological: on which space of composable classes is multiplication continuous? The quotient of composable representatives always works. Its topology need not agree with the ordinary subspace topology on pairs of quotient arrows, and product-quotient compatibility is exactly the condition under which the

two agree. An open arrow quotient guarantees this compatibility. For the universal presentation, the open-map theorem for locally path-connected, semilocally simply connected spaces is now checked directly in Lean. It yields an ordinary topological groupoid and discrete based fibers. For totally disconnected spaces the path-class projection itself is a homeomorphism, which gives the same composition conclusion directly. The Hawaiian earring shows that ordinary-product continuity on the observable based-fiber quotient does not extend to every locally path-connected space.

In the universal presentation every path is a primitive step and every endpoint-fixed homotopy is a rewrite, so its scoped quotient is the usual fundamental groupoid with the quotient topology on arrows. For a complete presentation, a continuous choice of trace for each represented path also identifies the trace-sensitive and observable quotient topologies. This applies to the universal presentation and the based-loop circle and torus examples. The complete discrete-label example over the harmonic archipelago shows that the trace choice cannot be dropped from this criterion. Section 10 distinguishes the Lean checks from the mathematical arguments and the immutable registered result.

The substitution theorem explains how this semantics behaves under changes of presentation. Primitive word codes supply the continuity conditions; scoped round trips supply the inverse maps on quotient classes. The resulting homeomorphisms cover the trace-sensitive arrow quotients and their final composable-pair domains. A conservative redundant-generator move can change the observable topology, and the checked timing example separates equal-slot from binary observation. These examples make the retained computational and parametrization data part of the mathematical question.

FUNDING

No external funding was received for this work.

ACKNOWLEDGEMENTS

AI-assisted tools were used during manuscript preparation for editorial support and assistance with software, documentation, and figure preparation. The geometric exposition and figures were developed from Tiago M. L. de Veras’s suggestions on reversal, coherence, and homotopy obstructions. The figures are mathematical schematics with explicitly stated maps and boundary conditions. They illustrate the arguments and are not offered as evidence for the theorems.

APPENDIX A. DECLARATION INDEX FOR THE PARENT ARTIFACT

This index records the representative declarations discussed in Section 10.1. It describes the parent Lean artifact, not the independently registered Palomar selection of Section 10.2. Fully qualified names are retained for source lookup.

Table 6: Representative declarations in the parent artifact, not the Palomar selection.

Mathematical layer	Representative checked declarations
Scoped syntax and soundness	<code>ScopedGeometricRewritePresentation</code> , <code>ScopedRwEq</code> , <code>ScopedRwEq.sound</code>

Table 6 continued

Mathematical layer	Representative checked declarations
Weighted realization	<code>WeightedSlotReparam</code> , <code>balancedSlotReparam</code> , <code>weightedConcatenationInvariant</code>
Trace-sensitive topology	<code>TraceSensitiveTopologicalCompPath.lean</code> , <code>continuous_fullTraceRealization</code> , <code>continuous_traceSensitive_to_observable</code> , <code>traceSensitiveTopologyCertificate</code> , <code>TraceSensitiveQuotient.continuous_quotientComparison</code> , <code>TraceSensitiveQuotient.quotientComparisonHomeomorph_of_realization_section</code>
Finite topology separation	<code>TraceSensitiveSeparation.oneLetter_e_ne_f</code> , <code>TraceSensitiveSeparation.observableCode_e_eq_f</code> , <code>TraceSensitiveSeparation.not_continuous_observable_to_trace</code> , <code>TraceSensitiveSeparation.certificate</code>
Quotient groupoid and topology	<code>ScopedComposableClass</code> , <code>scopedCompositionOnStrong</code> , <code>scopedFinalTopologicalGroupoidCertificate</code>
Ordinary/final comparison	<code>scopedProductCompatibility_iff_four_way</code> , <code>scopedProductCompatibility_of_compact_final_t2</code>
Comparison and normal forms	<code>ComparisonFunctorCertificate</code> , <code>comparisonHomeomorph_of_complete</code> , <code>ScopedGeometricNormalFormCertificate</code>
Universal and obstruction cases	<code>universalHomeomorph</code> , <code>ScopedGeometricRewrite</code> , <code>universalTraceSensitiveHomeomorph</code> , <code>QuotientObstructionTransfer</code> , <code>MultiplicationComparison</code>
Finite-generator presentation data	<code>FiniteCircleTorusPresentation</code> , <code>circleFinitePresentation</code> , <code>FiniteCircleTorusPresentation</code> , <code>circleCompletionEquivInt</code> , <code>FiniteCircleTorusPresentation</code> , <code>torusFinitePresentation</code> , <code>FiniteCircleTorusPresentation</code> , <code>torusCompletionEquivIntProd</code> , <code>FiniteCircleTorusPresentation</code> , <code>torusFiniteCommutingSquare</code>
Completed circle and topological torus	<code>circleBasedNormalFormCertificate</code> , <code>TopologicalTorus</code> , <code>standardLoop_homotopic_sequentialLoop</code> , <code>TopologicalTorus.equivIntProd</code> , <code>TopologicalTorus.certificate</code>

REFERENCES

1. Jeremy Brazas, *The fundamental group as a topological group*, *Topology and its Applications* **160** (2013), 170–188.
2. Jeremy Brazas and Paul Fabel, *On fundamental groups with the quotient topology*, *Journal of Homotopy and Related Structures* **10** (2015), 71–91.
3. Ronald Brown, *Topology and groupoids*, 3rd ed., BookSurge LLC, Charleston, SC, 2006.
4. Ronald Brown and Gholamreza Danesh-Naruie, *The fundamental groupoid as a topological groupoid*, *Proceedings of the Edinburgh Mathematical Society* **19** (1975), no. 3, 237–244.
5. Alonzo Church and J. Barkley Rosser, *Some properties of conversion*, *Transactions of the American Mathematical Society* **39** (1936), no. 3, 472–482.

6. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Proceedings of the 28th International Conference on Automated Deduction (CADE-28), Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
7. Ruy J. G. B. de Queiroz, Anjolina Grisi de Oliveira, and Arthur Freitas Ramos, *Propositional equality, identity types, and computational paths*, South American Journal of Logic **2** (2016), no. 2, 245–296.
8. Tiago M. L. de Veras, Arthur F. Ramos, Ruy J. G. B. de Queiroz, and Anjolina G. de Oliveira, *A topological application of labelled natural deduction*, South American Journal of Logic, Advance Access, forthcoming.
9. ———, *Calculation of fundamental groups via computational paths*, Anais do VI Encontro de Teoria da Computação (ETC 2021), Sociedade Brasileira de Computação – SBC, July 2021, pp. 17–21.
10. ———, *Computational paths – a weak groupoid*, Journal of Logic and Computation **35** (2025), no. 5, exad071, Published online 24 November 2023; issue dated 2025.
11. Paul Fabel, *The fundamental group of the harmonic archipelago*, 2005.
12. ———, *Multiplication is discontinuous in the Hawaiian Earring Group (with the quotient topology)*, Bulletin of the Polish Academy of Sciences. Mathematics **59** (2011), no. 1, 77–83.
13. Yves Guiraud and Philippe Malbos, *Higher-dimensional categories with finite derivation type*, Theory and Applications of Categories **22** (2009), no. 18, 420–478.
14. J. P. L. Hardy and Sidney A. Morris, *A note on free topological groupoids*, Mathematische Nachrichten **77** (1977), no. 1, 177–180.
15. Allen Hatcher, *Algebraic topology*, Cambridge University Press, Cambridge, 2002.
16. Simon Henry and Samuel Mimram, *Tietze equivalences as weak equivalences*, 2021.
17. Martin Hofmann and Thomas Streicher, *The groupoid interpretation of type theory*, Twenty-five Years of Constructive Type Theory (Giovanni Sambin and Jan M. Smith, eds.), Oxford Logic Guides, vol. 36, Oxford University Press, 1998, pp. 83–111.
18. Rohit Dilip Holkar, Md Amir Hossain, and Dheeraj Kulkarni, *The fundamental groupoid: Topology, haar system and actions*, Münster Journal of Mathematics **17** (2024), 203–240.
19. Donald E. Knuth and Peter B. Bendix, *Simple word problems in universal algebras*, Computational Problems in Abstract Algebra (J. Leech, ed.), Pergamon Press, Oxford, 1970, pp. 263–297.
20. J. Peter May, *A concise course in algebraic topology*, Chicago Lectures in Mathematics, University of Chicago Press, 1999.
21. Ernest Michael, *Bi-quotient maps and cartesian products of quotient maps*, Annales de l’Institut Fourier **18** (1968), no. 2, 287–302.
22. M. H. A. Newman, *On theories with a combinatorial definition of “equivalence”*, Annals of Mathematics **43** (1942), no. 2, 223–243.
23. Ali Pakdaman and Fereshteh Shahini, *The fundamental groupoid as a topological groupoid: Lasso topology, Topology and its Applications* **302** (2021), 107838.
24. ———, *Topological fundamental groupoids: Brown’s topology*, Hacettepe Journal of Mathematics and Statistics **52** (2023), no. 4, 896–906.
25. Arthur F. Ramos, Anjolina G. de Oliveira, Ruy J. G. B. de Queiroz, and Tiago M. L. de Veras, *Formalizing computational paths and fundamental groups in Lean*, 2025, arXiv preprint arXiv:2511.19142.
26. Arthur F. Ramos, Ruy J. G. B. de Queiroz, and Anjolina G. de Oliveira, *Computational paths and the fundamental groupoid of a type*, Anais do VI Encontro de Teoria da Computação (ETC 2021), Sociedade Brasileira de Computação – SBC, July 2021, pp. 22–25.
27. Arthur F. Ramos, Ruy J. G. B. de Queiroz, Anjolina G. de Oliveira, and Tiago M. L. de Veras, *Explicit computational paths*, South American Journal of Logic **4** (2018), no. 2, 441–484.
28. Arthur F. Ramos, Tiago M. L. de Veras, Ruy J. G. B. de Queiroz, and Anjolina G. de Oliveira, *Computational paths form a weak ω -groupoid*, 2025, arXiv preprint arXiv:2512.00657.
29. ———, *The Seifert–van Kampen theorem via computational paths: A formalized approach to computing fundamental groups*, 2025, arXiv preprint arXiv:2512.03175.
30. Arthur Freitas Ramos, Ruy J. G. B. de Queiroz, and Anjolina G. de Oliveira, *On the identity type as the type of computational paths*, Logic Journal of the IGPL **25** (2017), no. 4, 562–584.
31. Arthur Freitas Ramos, Ruy J. G. B. de Queiroz, Anjolina Grisi de Oliveira, and Tiago M. L. de Veras, *Topological semantics for scoped computational paths—lean artifact*, Zenodo software archive, 2026, Version 0.6.1+lean-only-v3, doi:10.5281/zenodo.21938980.

32. Arthur Freitas Ramos, Ruy J. G. B. de Queiroz, Anjolina Grisi de Oliveira, Tiago M. L. de Veras, and David Barros Hulak, *TopologicalComputationalPaths: registered lean result*, Palomar, PALOMAR-2026-08-29-000005, version 1, 2026, Registered 29 August 2026. Source commit 8254c40d0de03ff469c7c9c05087b8bc154e9a87.
33. Arthur Freitas Ramos, Ruy J. G. B. de Queiroz, Anjolina Grisi de Oliveira, and Dov M. Gabbay, *Computational paths: The calculus of equality*, College Publications, 2026.
34. Craig C. Squier, *Word problems and a homological finiteness condition for monoids*, Journal of Pure and Applied Algebra **49** (1987), no. 1–2, 201–217.